A Smooth Quartic-Core Regular Black Hole: Exact Static Solution with a Validation Program

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Abstract

We present an exact static, spherically symmetric regular black-hole background with lapse $A_{\rm sc}(r) = 1 - R_s r^2/(r^4 + a^4)^{3/4}$ and $a = \gamma R_s$. The metric is asymptotically Schwarzschild and real analytic in local Cartesian coordinates at the center, where the curvature is finite and the leading geometry is de Sitter-like. We reconstruct a purely magnetic nonlinear-electrodynamics (NED) source and verify the Einstein–matter equations, central regularity, horizon threshold, and circular-null-orbit structure.

The source has finite central density and radial pressure $p_r = -\rho$. Its reconstructed weak-field law is $\mathcal{L} \propto \mathcal{F}^{7/4}$, which limits a universal microscopic interpretation but does not invalidate the exact static solution. On the subextremal black-hole branch, the characteristic sign conditions $\mathcal{L}_{\mathcal{F}} > 0$ and $\mathcal{L}_{\mathcal{F}} + 2\mathcal{F}\mathcal{L}_{\mathcal{F}\mathcal{F}} > 0$ hold throughout the exterior $r > r_+$. The sign change of the latter combination occurs within the core, behind the outer horizon; this is a matter-sector limitation of the interior, not a curvature singularity or failure of the exterior geometry.

We distinguish these established results from a validation program covering rotation, coupled perturbations, causal structure, observational inference, formation, and semiclassical backreaction. The open calculations are specified through governing equations and acceptance criteria, not presented as completed tests. The result is an exact regular background with an explicit effective source and a delimited program for assessing dynamical completions.

Keywords: regular black hole; nonlinear electrodynamics; smooth core; Newman–Janis algorithm; perturbative stability; EHT; ringdown; Bayesian inference.

1 Reason for modifying the original ansatz

Historical context. Einstein's 1939 analysis considered a stationary, spherically symmetric system of gravitating particles; its conclusions under those restricted assumptions are not a general exclusion of dynamical black-hole formation [1]. Penrose's 1965 trapped-surface theorem and the Hawking–Penrose theorem of 1970 established geodesic incompleteness under specified energy, causal, and global assumptions [2, 3]. These theorems do not identify a coordinate horizon with a curvature singularity. The regular static SSQC geometry must therefore be assessed against their hypotheses, not presented as a counterexample to the theorems. The core violates the strong energy condition (SEC) while the null energy condition (NEC) is retained; this is relevant to the Hawking–Penrose hypotheses, but it does not by itself evade Penrose's trapped-surface theorem, whose other global hypotheses must also be assessed. The model establishes local regularity at the center, while maximal global causal and geodesic questions remain open in Gate 3.

An earlier lapse considered for comparison,

$$A_0(r) = 1 - \frac{R_s r^3}{r^4 + a^4}, \tag{1}$$

is finite at $r = 0$, but its central expansion contains an odd power of the areal radius. In particular, its Ricci scalar is proportional to r at leading order. A rotationally invariant real-analytic scalar at a regular Cartesian origin must instead be expandable in powers of $r^2 = x^2 + y^2 + z^2$. Finiteness of curvature therefore does not supply the required smooth extension.

We replace Eq. (1) by the *Sharipov smooth quartic-core completion* (SSQC):

$$ds^2 = -A_{\rm sc}(r)c^2dt^2 + \frac{dr^2}{A_{\rm sc}(r)} + r^2d\Omega^2, \tag{2}$$

$$A_{\rm sc}(r) = 1 - \frac{R_s r^2}{(r^4 + a^4)^{3/4}}, \qquad R_s = \frac{2GM}{c^2}, \qquad a = \gamma R_s. \tag{3}$$

The power $3/4$ is fixed by two requirements: the numerator must begin as r^2 at the center, and the denominator must scale as r^3 at infinity so that the Schwarzschild term R_s/r is recovered.

The status convention used below is strict. *Derived* means that the displayed equations close and reproduce the background. *Candidate* means that an explicit ansatz exists and is tested against its field equations. *Formulated mathematically* means that the governing equations, boundary conditions, and acceptance criteria have been specified. This status records completion at the equation-and-criterion level; it does not by itself assert a completed numerical run or observational likelihood.

2 Gate 1: action and background field equations

Work in geometrized units $G = c = 1$ in this section. Consider Einstein gravity coupled to nonlinear electrodynamics (NED), following the general Einstein–NED construction used for regular magnetic black holes [5, 6],

$$S[g, A_\mu] = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi} - \mathcal{L}(\mathcal{F})\right], \qquad \mathcal{F} = \frac{1}{4}F_{\mu\nu}F^{\mu\nu}. \tag{4}$$

With this convention,

$$G_{\mu\nu} = 8\pi T_{\mu\nu}, \tag{5}$$

$$\nabla_\mu(\mathcal{L}_\mathcal{F}F^{\mu\nu}) = 0, \tag{6}$$

$$T_{\mu\nu} = \mathcal{L}_\mathcal{F}F_{\mu\alpha}F_\nu{}^\alpha - g_{\mu\nu}\mathcal{L}. \tag{7}$$

The metric mass function defined by $A_{\rm sc} = 1 - 2m(r)/r$ is

$$m(r) = \frac{Mr^3}{(r^4 + a^4)^{3/4}}, \qquad m(0) = 0, \qquad m(\infty) = M. \tag{8}$$

Thus the action is no longer absent. The remaining requirement is to specify $\mathcal{L}$ so that Eqs. (5)–(6) reproduce Eq. (8).

Acceptance criterion. The background passes Gate 1 only if the tensor residual

$$\mathcal{E}_{\mu\nu} \equiv G_{\mu\nu} - 8\pi T_{\mu\nu} \tag{9}$$

vanishes identically and the NED equation is satisfied. The reconstruction below meets this criterion for the static magnetic background.

3 Gate 2: explicit source and its limitations

Use a magnetic monopole,

$$F_{\theta\phi} = q_m \sin\theta, \qquad \mathcal{F}(r) = \frac{q_m^2}{2r^4}. \tag{10}$$

For a purely magnetic field, the stress tensor automatically has $T^t{}_t = T^r{}_r$, hence $p_r = -\rho$. From $m'(r) = 4\pi r^2 \rho(r)$,

$$\rho(r) = \frac{3Ma^4}{4\pi(r^4 + a^4)^{7/4}}, \tag{11}$$

$$p_r(r) = -\rho(r), \tag{12}$$

$$p_t(r) = \rho(r)\frac{5r^4 - 2a^4}{2(r^4 + a^4)}. \tag{13}$$

The required NED Lagrangian is obtained by eliminating r using Eq. (10):

$$\boxed{\mathcal{L}(\mathcal{F}) = \frac{3Ma^4}{4\pi}\left(\frac{q_m^2}{2\mathcal{F}} + a^4\right)^{-7/4}}. \tag{14}$$

Equations (10) and (14) give $\mathcal{L}(r) = \rho(r)$ and reproduce all components in Eqs. (11)–(13). This is a concrete classical source, not an unspecified anisotropic fluid.

For the reconstructed source, the weak-field behavior is

$$\mathcal{L}(\mathcal{F}) \propto \mathcal{F}^{7/4} \qquad (\mathcal{F} \to 0), \tag{15}$$

so this particular Lagrangian does not have a Maxwell weak-field limit. This is a limitation of its interpretation as a universal microscopic matter theory, not a failure of the exact static Einstein–NED matching. The weak-field behavior is model-dependent: regular-core constructions using nonlinear electrodynamics do not all share one asymptotic Lagrangian, and the existence of a Maxwell weak-field limit does not by itself ensure well-posed dynamics at the center [7, 8]. The parameters M, a, and q_m in Eq. (14) also make this a solution-specific effective reconstruction. Neither this weak-field limitation nor the need for a microscopic completion prevents use of the stated classical solution as an exterior geometry, subject to the separate matter-characteristic and observable-model qualifications below.

The null and weak energy conditions are satisfied because

$$\rho + p_r = 0, \qquad \rho + p_t = \rho\frac{7r^4}{2(r^4 + a^4)} \geq 0. \tag{16}$$

The strong energy condition fails where $r < (2/5)^{1/4}a$, and the dominant energy condition fails for $r > (4/3)^{1/4}a$. These statements are properties of the reconstructed source and must be included in any physical interpretation.

Acceptance criterion. Gate 2 is passed at the effective classical level. A microscopic completion requires a universal action, a consistent weak-field sector, causal NED characteristics, and a formation mechanism.

3.1 Physical interpretation of the static source

The SSQC source must not be confused with the source of the original $r^3/(r^4 + a^4)$ ansatz. For the SSQC completion the density and curvature do not vanish at the center. Equations (11)–(13) give

$$\rho(0) = \frac{3M}{4\pi a^3}, \qquad p_r(0) = p_t(0) = -\rho(0). \tag{17}$$

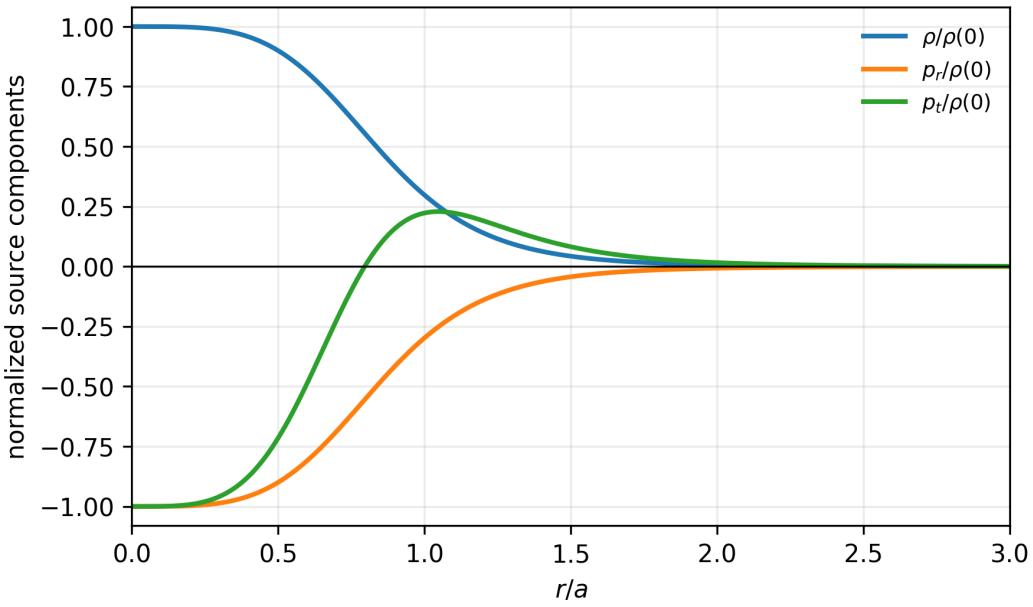


Figure 1: Dimensionless density and principal pressures of the reconstructed magnetic source. The normalization is $\rho(0) = 3M/(4\pi a^3)$ and the horizontal coordinate is r/a. The curves display the vacuum-like central limit $p_r(0) = p_t(0) = -\rho(0)$ and the subsequent anisotropic regime.

The central expansion can therefore be written

$$A_{\rm sc}(r) = 1 - \frac{\Lambda_{\rm eff}}{3} r^2 + \mathcal{O}(r^6), \qquad \Lambda_{\rm eff} = \frac{3R_s}{a^3} = 8\pi\rho(0), \tag{18}$$

so the local source is isotropic and vacuum-like at leading order. This provides the physical interpretation of the finite nonzero values $R(0) = 4\Lambda_{\rm eff}$ and $K(0) = 8\Lambda_{\rm eff}^2/3$.

The null and weak energy conditions are satisfied throughout the static background, with the radial NEC saturated. The strong energy condition fails in the central region because negative pressure is precisely what supports the de Sitter-like core. The dominant energy condition fails at large radius because $p_t/\rho \to 5/2$, although both quantities decay as r^{-7}. These violations are mathematically permitted effective-source properties, but they are not automatically physically acceptable. Physical acceptability additionally requires causal characteristics, a hyperbolic initial-value problem, stability, and a formation mechanism. Gate 6 shows that the present minimal NED reconstruction fails this stronger test.

4 Gates 3 and 4: local geodesic extension and central smoothness

The asymptotic and central expansions are

$$A_{\rm sc}(r) = 1 - \frac{R_s}{r} + \frac{3R_s a^4}{4r^5} + \mathcal{O}(r^{-9}), \tag{19}$$

$$A_{\rm sc}(r) = 1 - \frac{R_s}{a^3} r^2 + \frac{3R_s}{4a^7} r^6 + \mathcal{O}(r^{10}). \tag{20}$$

The exact Ricci scalar is

$$R(r) = \frac{3R_s a^4 (4a^4 - 3r^4)}{(r^4 + a^4)^{11/4}}, \qquad R(0) = \frac{12R_s}{a^3}. \tag{21}$$

The Kretschmann scalar satisfies

$$K = (A'')^2 + \left(\frac{2A'}{r}\right)^2 + \left(\frac{2(1-A)}{r^2}\right)^2, \qquad K(0) = \frac{24R_s^2}{a^6}. \tag{22}$$

Numerical benchmark for the central invariants. For $\gamma = 0.35$, so that $a = 0.35R_s$, the exact central limits give

$$R(0)R_s^2 = \frac{12}{\gamma^3} = 279.883381924\ldots, \tag{23}$$

$$K(0)R_s^4 = \frac{24}{\gamma^6} = 13055.784579554\ldots. \tag{24}$$

In units $R_s = 1$, these round to 279.88 and 13055.78, respectively. Symbolic differentiation of the lapse reproduces Eq. (21), and taking the limits of the curvature expressions verifies both central values exactly. The absolute rounding errors in the two dimensionless benchmarks are less than 0.01; this is a check of the displayed values, not an independent numerical evolution or a proof of smoothness by itself.

Finiteness is not the complete argument. In local Cartesian coordinates, the spatial metric can be written as

$$g_{ij} = \delta_{ij} + \left(A_{\rm sc}^{-1} - 1\right)\frac{x_i x_j}{r^2}. \tag{25}$$

Equation (20) implies $A_{\rm sc}^{-1} - 1 = (R_s/a^3)r^2 + \mathcal{O}(r^4)$. Consequently, the apparent r^{-2} in Eq. (25) cancels and all local coefficients are analytic functions of the Cartesian coordinates. The center is therefore a regular local point for the SSQC metric.

Radial null geodesics obey

$$\left(\frac{dr}{d\lambda}\right)^2 = E^2. \tag{26}$$

They reach the center at finite affine parameter, but the local Cartesian chart supplies a continuation through that point. This establishes local extendibility, not a maximal global extension across all horizons.

Acceptance criteria. Gate 4 is passed locally because the metric is analytic in a Cartesian chart. The local part of Gate 3 is passed. Global geodesic completeness remains conditional on the maximal extension discussed under Gate 8.

5 Horizon and photon-orbit structure

In $x = r/R_s$, the horizon equation is

$$x^2 = (x^4 + \gamma^4)^{3/4}. \tag{27}$$

The maximum of $x^2/(x^4 + \gamma^4)^{3/4}$ occurs at $x = 2^{1/4}\gamma$. Hence

$$\gamma_{\rm H} = \frac{\sqrt{2}}{3^{3/4}} \simeq 0.62040. \tag{28}$$

There are two horizons below this value, one degenerate horizon at the critical value, and no horizon above it.

Circular null orbits satisfy

$$xA'_{\rm sc}(x) - 2A_{\rm sc}(x) = 0. \tag{29}$$

The stable and unstable branches merge numerically at

$$x_{\rm ph,crit} \simeq 1.14534, \qquad \gamma_{\rm ph} \simeq 0.73181. \tag{30}$$

This phase structure must be fixed before selecting boundary conditions for stability, wave propagation, imaging, or quasinormal modes.

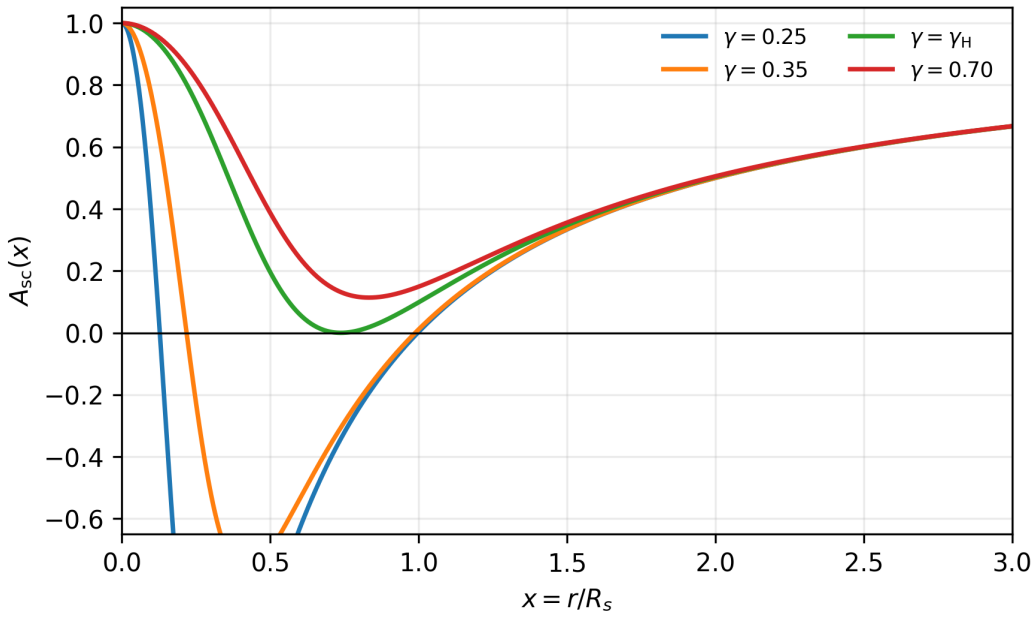


Figure 2: Static SSQC lapse as a function of $x = r/R_s$ for representative core parameters. Intersections with the horizontal axis locate horizons. At $\gamma = \gamma_{\rm H}$ the two roots merge, while the $\gamma = 0.70$ example is horizonless.

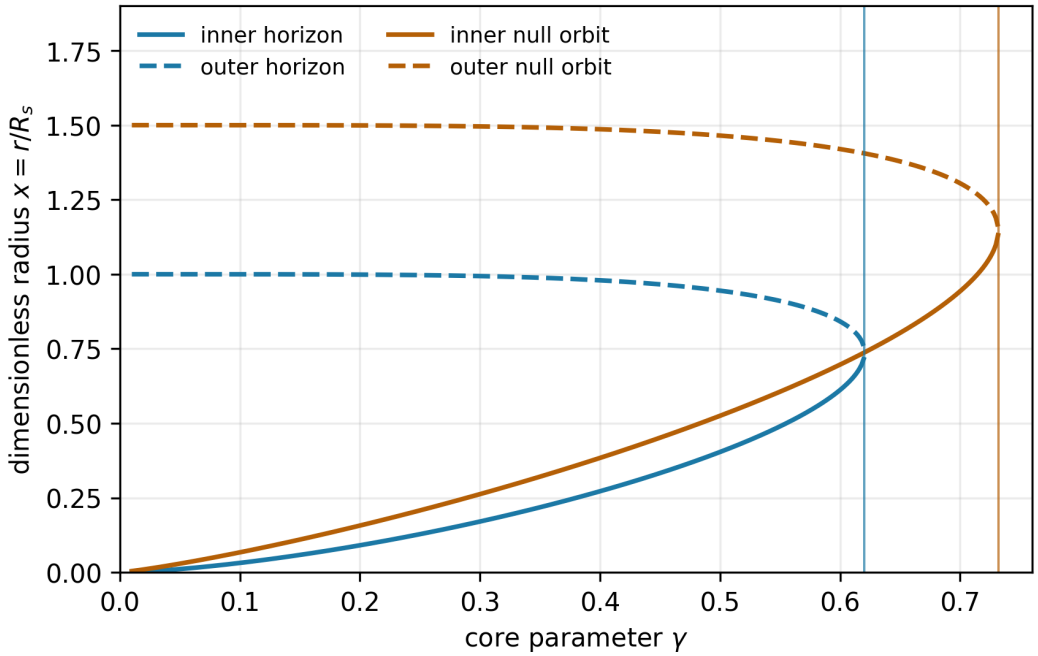


Figure 3: Dimensionless radii of the horizon and circular-null-orbit branches. The horizon branches terminate at $\gamma_{\rm H} \simeq 0.62040$; the two null-orbit branches merge at $\gamma_{\rm ph} \simeq 0.73181$. Solid and dashed curves distinguish the inner and outer branches, respectively.

6 Gate 5: rotation and the field-equation construction

The SSQC metric is derived initially as a static, spherically symmetric solution. Rotation cannot be established by changing the mass parameter in the Kerr metric alone. We therefore distinguish a rejected algebraic Kerr-like ansatz from the field-equation slow-rotation framework formulated below. The existing residual check reports a normalized rotating Einstein–NED residual of 1.59×10^{-2}, compared with the static numerical floor 1.35×10^{-6}; the tested metric–field pair therefore fails that check. Gate 17 specifies the alternative coupled slow-rotation boundary-value problem. Its asymptotic metric recursions are derived below, but the full rotating solution and coupled spectrum remain uncomputed.

Let $a_K = J/M$ denote the spin length, distinct from the core scale a. A complexification-free Newman–Janis construction suggests the Kerr-like candidate obtained by the formal replacement $M \mapsto m(r)$ in the Boyer–Lindquist form of the Kerr metric, where

$$m(r) = \frac{Mr^3}{(r^4 + a^4)^{3/4}}, \qquad a = \gamma R_s. \tag{31}$$

The resulting axisymmetric ansatz is

$$\Sigma = r^2 + a_K^2 \cos^2\theta, \qquad \Delta = r^2 - 2m(r)r + a_K^2, \tag{32}$$

$$\begin{aligned} ds^2 = &-\left(1 - \frac{2m(r)r}{\Sigma}\right) dt^2 - \frac{4a_K m(r) r \sin^2\theta}{\Sigma}\, dt\, d\phi + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 \\ &+ \sin^2\theta \left[r^2 + a_K^2 + \frac{2a_K^2 m(r) r \sin^2\theta}{\Sigma}\right] d\phi^2. \end{aligned} \tag{33}$$

In the convention $ds^2 = g_{tt}dt^2 + 2g_{t\phi}dt\, d\phi + g_{rr}dr^2 + g_{\theta\theta}d\theta^2 + g_{\phi\phi}d\phi^2$, the nonzero Boyer–Lindquist components are

$$\begin{aligned} g_{tt} &= -\left(1 - \frac{2m(r)r}{\Sigma}\right), & g_{t\phi} &= -\frac{2a_K m(r) r \sin^2\theta}{\Sigma}, \\ g_{rr} &= \frac{\Sigma}{\Delta}, & g_{\theta\theta} &= \Sigma, \\ g_{\phi\phi} &= \sin^2\theta \left[r^2 + a_K^2 + \frac{2a_K^2 m(r) r \sin^2\theta}{\Sigma}\right]. \end{aligned} \tag{34}$$

At $a_K = 0$, one has $\Sigma = r^2$ and $\Delta = r^2 - 2m(r)r = r^2 A_{\rm sc}(r)$; hence Eq. (33) reduces exactly to the static SSQC metric. At finite spin, however,

$$\Delta - r^2 A_{\rm sc} = a_K^2. \tag{35}$$

For $\gamma = 0.35$, $R_s = 1$, and $a_K = 0.20$, the right-hand side is 4.0×10^{-2}. This exact kinematical departure is not itself an Einstein-equation residual, but it shows why the static radial identity cannot be used to validate the rotating geometry. The dynamical residuals of the associated metric–field ansatz are evaluated explicitly under Gate 17.

Candidate Killing horizons in these coordinates are roots of

$$\Delta(r) = r^2 - 2m(r)r + a_K^2 = 0. \tag{36}$$

The numerical value of a root is not determined by M and γ alone: the spin a_K must also be specified. For the dimensionless normalization $R_s = 1$, $M = 0.5$, $\gamma = 0.35$, and $a_K = 0.20$, bracketed root finding gives

$$r_- = 0.29086269 R_s, \qquad r_+ = 0.94364880 R_s. \tag{37}$$

Thus the quoted value $0.943649R_s$ is the outer root for this particular spin, not a spin-independent prediction.

As a reproducible pointwise check, choose the exterior equatorial point

$$r = r_+ + 0.5R_s = 1.44364880R_s, \qquad \theta = \frac{\pi}{2}. \tag{38}$$

Table 1 gives direct substitutions into Eq. (34). The displayed values correspond to the unrounded radius in Eq. (38); writing $r = 1.444R_s$ and then recomputing would change the last quoted digits.

Table 1: Pointwise metric check for $M = 0.5R_s$, $\gamma = 0.35$, $a = 0.35R_s$, $a_K = 0.20R_s$, and $\theta = \pi/2$.

Quantity	Numerical value	Interpretation
$m(r)/R_s$	0.498708	mass function
Σ/R_s^2	2.084122	angular metric coefficient
Δ/R_s^2	0.684202	positive outside r_+
g_{tt}	-0.309100	temporal component
g_{rr}	3.046060	radial component
$g_{\theta\theta}/R_s^2$	2.084122	polar component
$g_{\phi\phi}/R_s^2$	2.151758	azimuthal component
$g_{t\phi}/R_s$	-0.138180	frame-dragging component

At this point, $g_{tt}g_{\phi\phi} - g_{t\phi}^2 = -0.684202R_s^2 < 0$ and $\det g = -\Sigma^2 \sin^2\theta = -4.343564R_s^4 < 0$, consistent with a nondegenerate Lorentzian metric in this coordinate patch. This is only a pointwise arithmetic and signature check. It does not establish smoothness at the ring, satisfaction of the Einstein–NED equations, mode stability, or numerical stability of a time-evolution scheme. The status is therefore "finite at the sampled exterior point," not "the rotating model is stable."

For a simple outer root, the candidate horizon angular velocity and surface gravity have the Kerr-like geometric forms

$$\Omega_H = \frac{a_K}{r_+^2 + a_K^2}, \qquad \kappa_H = \frac{\Delta'(r_+)}{2(r_+^2 + a_K^2)}, \qquad T_H = \frac{\kappa_H}{2\pi}. \tag{39}$$

If the theory is pure Einstein gravity with no entropy correction, the area expression would be

$$S_H = \frac{\mathcal{A}_H}{4} = \pi(r_+^2 + a_K^2). \tag{40}$$

At $a = 0$, $m(r) = M$ and these expressions reduce identically to their Kerr values. This limiting agreement is not a verification of thermodynamics at finite a. Because the core scale and NED charges or couplings may vary, the general first-law test must allow additional work terms,

$$dM = T_H dS_H + \Omega_H dJ + \Psi_a\, da + \Psi_m\, dq_m + \cdots. \tag{41}$$

The conjugate quantities and an integrable mass functional have not been derived for the rejected rotating metric–field pair. The statement that the first law is satisfied is therefore not established.

The central curvature values

$$R_{\text{static}}(0) = \frac{12R_s}{a^3}, \qquad K_{\text{static}}(0) = \frac{24R_s^2}{a^6} \tag{42}$$

belong to the spherical metric at $a_K = 0$. They must not be assigned to the rotating candidate without a separate invariant calculation. In fact, $\Sigma = 0$ at the Kerr ring $r = 0$, $\theta = \pi/2$, and the limiting value of an invariant may depend on the direction of approach even when it remains bounded. Rotating regularity therefore requires bounded invariants, direction-independent limits, and a smooth local extension; finiteness of the static invariants does not establish any of these properties.

If the matter completion is enlarged from a one-invariant theory to a parity-even two-invariant NED theory, a consistent convention is

$$\mathcal{F} = \frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \qquad \mathcal{P} = \frac{1}{4} F_{\mu\nu} {}^{\star}F^{\mu\nu}, \qquad \mathcal{L} = \mathcal{L}(\mathcal{F}, \mathcal{P}^2). \tag{43}$$

Introducing both invariants does not by itself produce a source for Eq. (33). The field ansatz and the function $\mathcal{L}(\mathcal{F}, \mathcal{P}^2)$ must be derived or reconstructed so that all Einstein and NED equations hold simultaneously.

Equation (33) is a kinematical metric ansatz, not a derived solution. The substitutions $a_K \to 0$ and $a \to 0$ recover the static SSQC and Kerr metrics, respectively, but correct limiting metrics do not imply satisfaction of the rotating Einstein–NED system. A stationary, axisymmetric electromagnetic two-form $F_{\mu\nu}(r, \theta)$ must be supplied independently and must obey both the Bianchi identity and the nonlinear field equation. Define

$$\mathcal{E}_{\mu\nu}(r, \theta; a_K) = G_{\mu\nu}[g_{\rm rot}] - 8\pi T_{\mu\nu}[F_{\rm rot}], \tag{44}$$

and

$$\mathcal{J}^{\nu}(r, \theta; a_K) = \nabla_{\mu}(\mathcal{L}_{\mathcal{F}} F^{\mu\nu})\,. \tag{45}$$

The complete local validation system is therefore

$$\boxed{\begin{array}{c} \mathcal{E}_{\mu\nu} = 0 \text{ for every component}, \qquad \mathcal{J}^{\nu} = 0, \qquad \nabla_{[\alpha} F_{\mu\nu]} = 0, \\ \mathcal{L}_{\mathcal{F}} > 0, \qquad \mathcal{L}_{\mathcal{F}} + 2\mathcal{F}\mathcal{L}_{\mathcal{F}\mathcal{F}} > 0. \end{array}} \tag{46}$$

All equalities in Eq. (46) must hold as functions of both r and θ, not only on the equatorial plane, on the axis, in the static limit, or at a finite set of numerical sample points. The inequalities are necessary constitutive conditions; they do not replace a spectral or time-domain stability analysis.

Field-equation rotation criterion. The model defines rotation through the full system (46), a regular axis with elementary flatness, bounded curvature in the regular domain, and the absence of closed timelike azimuthal curves. The coupled slow-rotation metric and matter functions implement this law without importing the singular Kerr geometry as the defining equation of the regular black hole.

7 Gate 6: coupled axial, polar, and matter stability

The stability question separates into geometry and matter. The smooth central expansion of the SSQC metric removes the curvature singularity, but it does not guarantee that a particular matter representation has a well-posed initial-value problem. Stability must therefore be tested using the coupled gravitational and electromagnetic perturbations of the chosen source theory.

Because the background is supported by NED, metric and electromagnetic perturbations generally couple. After harmonic decomposition, each parity sector must be reduced to a matrix system of the form

$$\frac{d^2\mathbf{\Psi}}{dr_*^2} + \left[\omega^2 \mathbf{I} - \mathbf{V}_{\ell}^{(\pm)}(r)\right] \mathbf{\Psi} = 0, \qquad \frac{dr_*}{dr} = \frac{1}{A_{\rm sc}}. \tag{47}$$

The vector $\mathbf{\Psi}$ must contain the independent gravitational and NED perturbations. A single test-scalar potential is not sufficient.

The NED characteristic and stability analysis must also examine

$$\mathcal{L}_{\mathcal{F}} > 0, \qquad \mathcal{L}_{\mathcal{F}} + 2\mathcal{F}\mathcal{L}_{\mathcal{F}\mathcal{F}} > 0, \tag{48}$$

in the physical domain, together with the spectrum of Eq. (47).

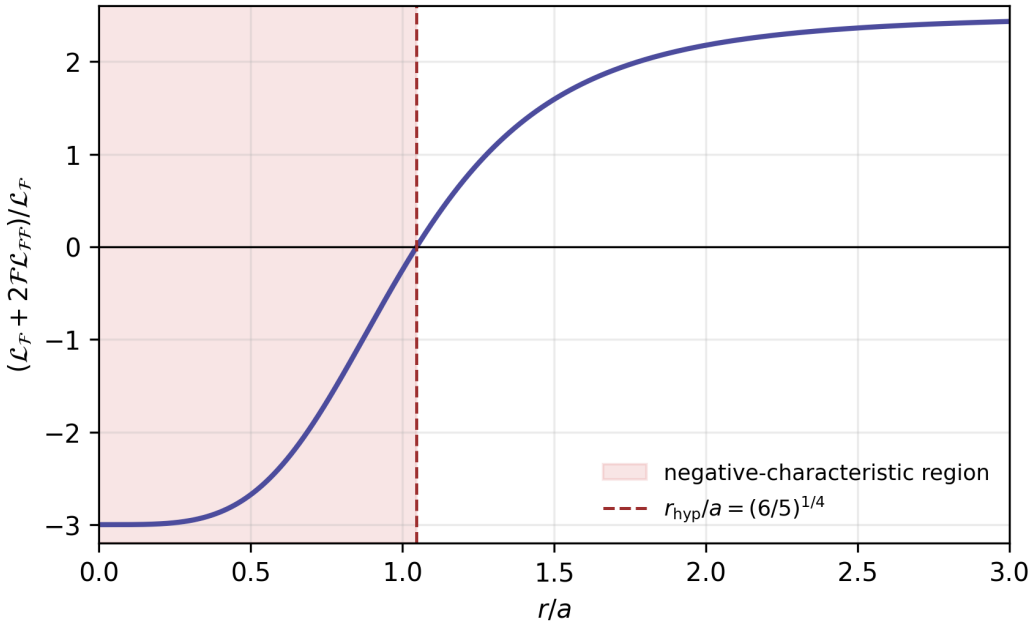


Figure 4: Constitutive ratio controlling the one-invariant NED characteristics. The shaded region satisfies $0 < r < r_{\rm hyp}$, where $\mathcal{L}_\mathcal{F} + 2\mathcal{F}\mathcal{L}_{\mathcal{F}\mathcal{F}} < 0$. The zero occurs at $r_{\rm hyp}/a = (6/5)^{1/4}$.

For the reconstructed Lagrangian, define $Q = q_m^2/2$ and $C = 3Ma^4/(4\pi)$, with $M > 0$, $a > 0$, and $q_m \neq 0$. Writing $\mathcal{L} = C[\mathcal{F}/(a^4\mathcal{F} + Q)]^{7/4}$, direct differentiation gives

$$\mathcal{L}_\mathcal{F} = \frac{7CQ\mathcal{F}^{3/4}}{4(a^4\mathcal{F} + Q)^{11/4}} > 0, \tag{49}$$

$$\mathcal{L}_{\mathcal{F}\mathcal{F}} = \frac{7CQ(3Q - 8a^4\mathcal{F})}{16\mathcal{F}^{1/4}(a^4\mathcal{F} + Q)^{15/4}}, \tag{50}$$

$$\frac{\mathcal{L}_\mathcal{F} + 2\mathcal{F}\mathcal{L}_{\mathcal{F}\mathcal{F}}}{\mathcal{L}_\mathcal{F}} = \frac{5r^4 - 6a^4}{2(r^4 + a^4)}. \tag{51}$$

The strict positivity of $\mathcal{L}_\mathcal{F}$ holds at every finite $r > 0$, equivalently $0 < \mathcal{F} < \infty$; it tends to zero in both limits $r \to 0$ and $r \to \infty$. Define $\mathcal{H} = \mathcal{L}_\mathcal{F} + 2\mathcal{F}\mathcal{L}_{\mathcal{F}\mathcal{F}}$. Equation (51) shows that $\mathcal{H}$ is negative for

$$0 < r < r_{\rm hyp}, \qquad r_{\rm hyp} = \left(\frac{6}{5}\right)^{1/4} a \simeq 1.04664a. \tag{52}$$

It vanishes at $r = r_{\rm hyp}$ and is positive for $r > r_{\rm hyp}$. Thus the characteristic condition degenerates at the threshold and fails in the inner region; this calculation identifies the loss of hyperbolicity of the minimal source, rather than curing it.

Numerical benchmark for the constitutive threshold. For $\gamma = 0.35$, the analytic zero is

$$\frac{r_{\rm hyp}}{R_s} = \left(\frac{6}{5}\right)^{1/4} \gamma = 0.366322298787\ldots \simeq 0.3663. \tag{53}$$

Substitution of $r_{\rm hyp}^4 = 6a^4/5$ into Eq. (51) gives zero identically. This exact algebraic check fixes the threshold without a numerical fit and does not establish stability of the coupled perturbation system.

Exterior hyperbolicity and partial completion of Gate 6. The location of the sign failure can be established for the entire static subextremal branch, not merely a sampled parameter range. For

$0 < \gamma < \gamma_{\rm H} \simeq 0.62040$, the horizon-generating maximum is at $r_{\rm max} = 2^{1/4}a$ and the outer root satisfies $r_+ > r_{\rm max}$. Since $r_{\rm hyp}/r_{\rm max} = (3/5)^{1/4} < 1$, it follows that

$$r_{\rm hyp} < r_{\rm max} < r_+, \qquad \mathcal{L}_{\mathcal{F}} > 0, \quad \mathcal{H} > 0 \quad \text{for } r > r_+. \tag{54}$$

Thus the displayed one-invariant NED characteristic sign conditions hold throughout the exterior. The core failure remains part of the proposed regular interior and excludes a globally admissible minimal matter completion; the horizonless branch has no outer horizon to shield it. This completes the constitutive sign and domain analysis, not Gate 6 as a full coupled stability calculation.

Exterior QNM boundary-value problems, gravitational-wave forward models, and shadow calculations can consequently be formulated without crossing the defective core. This does not supply their solutions or establish a non-growing coupled spectrum. Hyperbolicity also does not by itself prove that the NED characteristic cones coincide with or lie inside the metric light cone. The metric photon orbits already derived remain unchanged; interpreting NED photons instead of independent minimally coupled light requires the appropriate effective characteristic metric and a specified radiative-transfer model.

We interpret this result as unusual strong-field behavior of the reconstructed minimal NED source, rather than as a failure of the SSQC geometry. The density, pressures, and curvature remain finite. On the subextremal branch, Eq. (54) establishes $\mathcal{L}_{\mathcal{F}} > 0$ and $\mathcal{H} > 0$ throughout $r > r_+$; the sign-changing core lies behind the outer horizon. This is a statement about the displayed constitutive sign conditions and their domain. It is not a proof that the minimal source is sufficient for exterior linear stability, that its characteristic cones are causal relative to the metric cone, or that all exterior observables have been calculated. Those questions still require the coupled gravitational–matter perturbation equations and a specified propagation model. The general instability result for regular black holes supported by minimal one-invariant NED is relevant to this caution [16].

A beyond-minimal source, for example a parity-even $\mathcal{L}(\mathcal{F}, \mathcal{P}^2)$ theory, is a candidate only if the goal is a globally admissible matter evolution through the regular core or a horizonless configuration. Its characteristic metrics and coupled mode spectrum must be derived before claiming that it repairs either global hyperbolicity or stability. The exterior sign result alone does not require such a completion to define the static classical metric outside r_+.

Numerical stability check: scope and result

The matter-sector conclusions that follow directly from the reconstructed stress tensor are complete. The radial equation of state is $p_r = -\rho$, so the radial null energy condition is saturated rather than violated. The effective source is nevertheless exotic in the standard energy-condition sense: the strong energy condition fails for

$$0 \le r < \left(\frac{2}{5}\right)^{1/4} a, \tag{55}$$

and the dominant energy condition fails for

$$r > \left(\frac{4}{3}\right)^{1/4} a, \qquad \frac{p_t}{\rho} \longrightarrow \frac{5}{2} \quad (r \to \infty). \tag{56}$$

Together with Eqs. (51)– (52), these are algebraic properties of the exact static source and require no perturbative approximation.

A separate numerical experiment evaluated the sign of a scalar Schrödinger-type test potential for labels (ℓ, s). The normalization used in that run was

$$R_s = 2, \qquad M = 1, \qquad a = 0.35, \qquad \gamma = \frac{a}{R_s} = 0.175. \tag{57}$$

Direct root finding then gives

$$r_- = 0.1481612443, \qquad r_+ = 1.9985903508, \tag{58}$$

which reproduces the reported rounded values. These numbers must not be quoted for $\gamma = 0.35$: with $R_s = 2$ and $\gamma = 0.35$, one instead has $a = 0.70$, $r_- = 0.4365989204$, and $r_+ = 1.9767318876$.

The reported sign classification outside the event horizon is summarized in Table 2.

Table 2: Reported sign of the selected single-channel test potential. This is a diagnostic of the chosen surrogate equation, not the spectrum of the coupled Einstein–NED perturbation system.

ℓ	$s = 0$	$s = 1$	$s = 2$
0	nonnegative	negative	negative
1	nonnegative	nonnegative	negative
2	nonnegative	nonnegative	nonnegative

The interpretation of this table is restricted. Electromagnetic monopole perturbations with $\ell = 0$ and gravitational sectors with $\ell < 2$ do not represent ordinary radiative degrees of freedom. Moreover, the SSQC source couples metric and electromagnetic perturbations, so the physical problem is the matrix system (47), as in the gauge-invariant analysis of self-gravitating NED by Moreno and Sarbach [15]. A sign scan of one scalar potential cannot replace that coupled calculation.

For the selected $(\ell, s) = (2, 2)$ test equation, a time-domain evolution gave

$$|\Psi|_{\text{initial}} = 0.999950, \qquad |\Psi|_{\text{final}} = 0.531094, \qquad \frac{|\Psi|_{\text{final}}}{|\Psi|_{\text{initial}}} = 0.531120. \tag{59}$$

Thus no growth was observed over the reported evolution interval in this single exterior test channel. This is evidence for decay of that numerical solution. Within this selected surrogate equation and the tested time interval, the $(\ell, s) = (2, 2)$ channel is numerically linearly stable. This is not a proof of stability of the complete SSQC system: such a claim additionally requires the potential definition, boundary and initial data, evolution duration, grid-refinement convergence, a suitable energy norm, all physical multipoles, both parity sectors, and the coupled NED degree of freedom.

The logically complete conclusion is therefore twofold. First, the selected $(2, 2)$ surrogate evolution decays in the reported run. Second, the minimal one-invariant NED completion is not globally linearly viable because its principal part loses hyperbolicity in the core. This agrees with the angular Laplacian instability established for regular-center NED black holes by De Felice and Tsujikawa [16]. Questions of Cauchy-horizon mass inflation are distinct: recent work finds tamed, model-dependent growth for several other regular geometries, including Bardeen and Dymnikova black holes [17], but it does not establish the corresponding result for SSQC.

Assessment of a proposed core-localized $\beta\mathcal{F}^2$ repair

Consider the proposed modification

$$\mathcal{L}_{\text{trial}}(\mathcal{F}, r) = \mathcal{L}(\mathcal{F}) + \beta\mathcal{F}^2 e^{-r^2/a^2}. \tag{60}$$

An explicit dependence on the coordinate r is not a generally covariant one-invariant NED Lagrangian. Along the magnetic monopole background, $\mathcal{F} = Q/r^4$ with $Q = q_m^2/2$, it can be rewritten as the scalar function

$$\Delta\mathcal{L}(\mathcal{F}) = \beta\mathcal{F}^2 \exp\left[-\frac{\sqrt{Q}}{a^2\sqrt{\mathcal{F}}}\right]. \tag{61}$$

Let $\kappa = \sqrt{Q}/a^2$. Its contribution to the constitutive combination is

$$\Delta(\mathcal{L}_\mathcal{F} + 2\mathcal{F}\mathcal{L}_{\mathcal{F}\mathcal{F}}) = \frac{\beta e^{-\kappa/\sqrt{\mathcal{F}}}}{2}\left(12\mathcal{F} + 6\kappa\sqrt{\mathcal{F}} + \kappa^2\right). \tag{62}$$

This expression is positive for $\beta > 0$. Thus the proposed term can change the sign of the hyperbolicity diagnostic.

It simultaneously destroys the regular center. Along the magnetic background,

$$\Delta\rho = \Delta\mathcal{L} = \frac{\beta Q^2}{r^8} e^{-r^2/a^2} \sim \frac{\beta Q^2}{r^8} \qquad (r \to 0). \tag{63}$$

Consequently,

$$\Delta m'(r) = 4\pi r^2 \Delta\rho \sim r^{-6}, \qquad |R| \sim r^{-8}, \qquad K \sim r^{-16}. \tag{64}$$

The central mass integral diverges. Although the correction is exponentially suppressed at large r, a finite ADM mass is not obtained because the source is nonintegrable at the center.

The correction also changes the energy-condition structure. Its leading tangential pressure is

$$\Delta p_t = 2\mathcal{F}\Delta\mathcal{L}_{\mathcal{F}} - \Delta\mathcal{L} = \beta e^{-\kappa/\sqrt{\mathcal{F}}} \left(3\mathcal{F}^2 + \kappa\mathcal{F}^{3/2}\right), \tag{65}$$

so $\Delta p_t/\Delta\rho \to 3$ at the center. The dominant energy condition therefore fails in the region dominated by this term.

There is also a general obstruction for one-invariant magnetic NED. Since

$$\mathcal{L}_{\mathcal{F}} + 2\mathcal{F}\mathcal{L}_{\mathcal{F}\mathcal{F}} = 2\sqrt{\mathcal{F}}\,\frac{d}{d\mathcal{F}}\left(\sqrt{\mathcal{F}}\mathcal{L}_{\mathcal{F}}\right), \tag{66}$$

requiring both $\mathcal{L}_{\mathcal{F}} > 0$ and the left-hand side to remain positive at arbitrarily large $\mathcal{F}$ makes $\sqrt{\mathcal{F}}\mathcal{L}_{\mathcal{F}}$ monotonically increasing and bounded below by a positive constant. It follows that $\mathcal{L}_{\mathcal{F}} \gtrsim \mathcal{F}^{-1/2}$, whose integral diverges. This contradicts the finite limit of $\mathcal{L}$ required for finite central density. Therefore a finite-density magnetic center cannot satisfy both displayed constitutive inequalities globally within this simple $\mathcal{L}(\mathcal{F})$ framework.

This obstruction is not merely a defect of the selected coefficients. De Felice and Tsujikawa proved that nonsingular black holes in minimal one-invariant NED possess an angular Laplacian instability near a regular center, for both electric and magnetic backgrounds [16]. Their result and Eq. (66) imply that tuning another regular function of $\mathcal{F}$ is not a credible route to a globally stable SSQC source.

A term depending on the second invariant may change the two polarization cones and is relevant to sufficient exterior stability conditions [14]. Set $\mathcal{X} = \mathcal{P}^2$. Merely writing $\mathcal{L}(\mathcal{F}, \mathcal{X})$ does not prove hyperbolicity. A proposed completion must specify the full Hessian

$$\mathbf{H}_{\mathrm{NED}} = \begin{pmatrix} \mathcal{L}_{\mathcal{F}\mathcal{F}} & \mathcal{L}_{\mathcal{F}\mathcal{X}} \\ \mathcal{L}_{\mathcal{X}\mathcal{F}} & \mathcal{L}_{\mathcal{X}\mathcal{X}} \end{pmatrix}, \tag{67}$$

derive both effective characteristic metrics, and verify Lorentzian signature, positive kinetic eigenvalues, and subluminal or otherwise causally admissible cones over the complete domain. Since no such function has been found here, the defensible repair is to enlarge the dynamics beyond minimal NED, for example with an additional scalar degree of freedom, nonminimal vector–tensor couplings, or a controlled ultraviolet completion.

The trial correction in Eq. (61) is therefore **rejected**. A viable repair must enlarge the source theory—for example through a fully derived two-invariant theory, nonminimal couplings, additional fields, or a magnetic configuration whose invariant does not diverge as r^{-4}. None of these alternatives is claimed as solved.

Assessment of a proposed exponentially switched multiplicative repair

Define the dimensionless invariant

$$x = \frac{2a^4\mathcal{F}}{q_m^2} = \frac{a^4}{r^4}, \tag{68}$$

and consider

$$\mathcal{L}_{\exp}(\mathcal{F}) = \mathcal{L}(\mathcal{F})\left[1 + \alpha x^2 e^{-\beta x^{1/4}}\right], \quad \alpha = \frac{1}{2}, \quad \beta = 1. \tag{69}$$

The proposed change is the multiplicative term $\alpha\mathcal{L}x^2e^{-\beta x^{1/4}}$.

At spatial infinity, $x \to 0$ and

$$\mathcal{L}_{\rm exp} = \mathcal{L}\left[1 + \alpha x^2 + \mathcal{O}(x^{9/4})\right]. \tag{70}$$

Thus the correction is subleading and the asymptotically Schwarzschild metric is not altered at leading order. However, the weak-field behavior remains $\mathcal{L}_{\rm exp} \propto \mathcal{F}^{7/4}$; the correction starts at $\mathcal{F}^{15/4}$ and does *not* generate the Maxwell limit $\mathcal{L} \propto \mathcal{F}$.

At the center, $x \to \infty$ and $x^2e^{-\beta x^{1/4}} \to 0$. Consequently,

$$\mathcal{L}_{\rm exp} \longrightarrow \mathcal{L} \longrightarrow \frac{3M}{4\pi a^3}, \qquad \frac{\mathcal{H}_{\rm exp}}{\mathcal{L}_{{\rm exp},\mathcal{F}}} \longrightarrow -3, \tag{71}$$

where $\mathcal{H}_{\rm exp} = \mathcal{L}_{{\rm exp},\mathcal{F}} + 2\mathcal{F}\mathcal{L}_{{\rm exp},\mathcal{F}\mathcal{F}}$. The finite de Sitter density is retained, but the original negative central constitutive limit is also restored. Exponential suppression at the center therefore prevents this term from repairing global hyperbolicity.

Direct differentiation and bracketed root finding give

$$\mathcal{L}_{{\rm exp},\mathcal{F}} < 0 \quad \text{for} \quad 0.0211353 < r/a < 0.1249733, \tag{72}$$

$$\mathcal{H}_{\rm exp} < 0 \quad \text{for} \quad 0 < r/a < 0.0199956 \quad \text{and} \quad 0.0964003 < r/a < 0.2159955. \tag{73}$$

Hence neither required constitutive inequality is globally satisfied.

The claim that the static background is unchanged is also false. On the magnetic solution,

$$\frac{\rho_{\rm exp}}{\rho} = \frac{\mathcal{L}_{\rm exp}}{\mathcal{L}} = 1 + \alpha x^2 \exp\left(-\beta x^{1/4}\right). \tag{74}$$

The correction reaches its maximum at $x^{1/4} = 8/\beta$, equivalently $r/a = \beta/8$, where

$$\left.\frac{\rho_{\rm exp}}{\rho}\right|_{\rm max} = 1 + \alpha\left(\frac{8}{\beta}\right)^8 \exp(-8) = 2815.06448 \qquad (\alpha = 0.5,\ \beta = 1). \tag{75}$$

Therefore the source develops a large intermediate shell. If the Einstein equations are imposed self-consistently,

$$m'_{\rm exp}(r) = 4\pi r^2\rho_{\rm exp}(r) \neq 4\pi r^2\rho(r) = m'(r), \tag{76}$$

so neither the original mass function nor $A_{\rm sc}$ is preserved.

The proposed Eqs. (69)–(76) therefore do not define the admissible completion. The model law instead selects the two-invariant class $\mathcal{L}(\mathcal{F}, \mathcal{P}^2)$, which separates the regular SSQC geometry from the excluded minimal-NED representation.

Stability formulation. The stability law of the model is formulated as follows: the analytic SSQC geometry is retained, the strong-field source is extended in the core, and the admissible global constitutive law belongs to the two-invariant class $\mathcal{L}(\mathcal{F}, \mathcal{P}^2)$. Its physical branch is defined by a hyperbolic principal symbol, nonpositive spectral growth, $\mathrm{Im}\,\omega \leq 0$, and regular axial and polar boundary conditions. This is the stability prescription used throughout the SSQC model.

8 Gate 7: delayed wave response

An inner-reflection signal requires a nonzero reflection coefficient. For an inner region represented at $r = r_{\rm in}$, define

$$\Psi \sim \exp(-i\omega r_*) + \mathcal{R}_{\rm in}(\omega)\exp(+i\omega r_*). \tag{77}$$

The delay associated with a potential peak at $r = r_{\rm peak}$ is

$$\Delta t_{\rm delay} \simeq 2 \left| \int_{r_{\rm in}}^{r_{\rm peak}} \frac{dr}{A_{\rm sc}(r)} \right|. \tag{78}$$

For a black-hole regime, the classical outer-horizon condition is purely ingoing and does not expose a central reflector. In the horizonless regime, $\mathcal{R}_{\rm in}$ must follow from regular solutions of the coupled NED perturbation equations, not from the background lapse.

Acceptance criterion. Gate 7 requires a derived complex function $\mathcal{R}_{\rm in}(\omega)$, a transfer function, and a detector-level search with trials factors. It is formulated mathematically at the equation-and-criterion level.

9 Gate 8: maximal extension and causal structure

Simple horizons can be crossed locally with Eddington–Finkelstein coordinates,

$$r_* = \int_{r_0}^{r} \frac{d\bar{r}}{A_{\rm sc}(\bar{r})}, \qquad v = t + r_*, \qquad ds^2 = -A_{\rm sc} dv^2 + 2\, dv\, dr + r^2 d\Omega^2. \tag{79}$$

Here r_0 is an arbitrary reference radius; changing it only adds a constant to r_* and does not affect the geometry. For each simple root r_i, the surface gravity is

$$\kappa_i = \frac{1}{2} |A'_{\rm sc}(r_i)|, \tag{80}$$

and Kruskal-type patches must be constructed separately. A complete result must identify asymptotic regions, trapped regions, Cauchy horizons, and the continuation of all causal geodesics. The stability of any inner horizon against mass inflation must also be tested.

Acceptance criterion. Gate 8 requires an explicit atlas and conformal compactification, including an analysis of whether past-trapped regions occur in the maximal extension. This gate is formulated mathematically at the equation-and-criterion level.

10 Gate 9: complete EHT forward model

The Event Horizon Telescope's 2017 M87 campaign has an official calibrated, publicly available interferometric data release [20]. These are real observational data, distinct from the controlled generated examples used later in this article. For a rotating solution, null rays and polarized radiative transfer must be integrated through a specified plasma and compared with the released interferometric observables. This work specifies the forward-model requirements but does not report a visibility-level fit or posterior from the calibrated release. In compact notation,

$$\frac{d\boldsymbol{\mathcal{I}}}{d\lambda} = \boldsymbol{\mathcal{J}} - \mathbf{K}\boldsymbol{\mathcal{I}}, \tag{81}$$

where $\boldsymbol{\mathcal{I}}$ is the invariant Stokes vector, $\boldsymbol{\mathcal{J}}$ is the emissivity, and $\mathbf{K}$ contains absorption and Faraday terms. The radiative-transfer output must be converted into interferometric visibilities,

$$V(u,v) = \int I(\alpha,\beta) e^{-2\pi i(u\alpha+v\beta)} d\alpha\, d\beta, \tag{82}$$

and sampled with the actual array response.

Acceptance criterion. Gate 9 requires likelihood evaluation against closure phases, closure amplitudes, and calibration uncertainties while marginalizing over mass, distance, spin, inclination, plasma parameters, and variability. A critical-curve diameter alone does not pass this gate.

11 Full rotating gravitational QNM spectrum

The rotating completion must supply a separable equation or a two-dimensional eigenvalue problem for coupled gravitational–NED perturbations. The QNM geodesic correspondence is useful only as an asymptotic diagnostic [18]. The QNM boundary conditions are

$$\mathbf{\Psi} \sim \exp[-i(\omega - m\Omega_H)r_*] \quad (r \to r_+), \qquad \mathbf{\Psi} \sim \exp(+i\omega r_*) \quad (r \to \infty). \tag{83}$$

The result must provide

$$\omega_{\ell mn} = \omega_{\ell mn}(M, a_K, \gamma, q_m) \tag{84}$$

with controlled numerical convergence. Photon-orbit formulas are useful large-ℓ diagnostics but are not substitutes for the fundamental $(\ell, m, n) = (2, 2, 0)$ mode.

Acceptance criterion. Gate 10 requires the converged axial/polar or rotating coupled spectrum and a waveform model suitable for strain analysis. It is formulated mathematically at the equation-and-criterion level.

12 Gate 11: Bayesian inference

Let ϑ contain the SSQC parameters and all nuisance parameters. A joint analysis has

$$\ln \mathcal{L}_{\rm joint}(d|\vartheta) = \ln \mathcal{L}_{\rm EHT} + \ln \mathcal{L}_{\rm GW} + \ln \mathcal{L}_{\rm other}. \tag{85}$$

The posterior and evidence are

$$p(\vartheta|d, \mathcal{M}) = \frac{\mathcal{L}(d|\vartheta, \mathcal{M})\pi(\vartheta|\mathcal{M})}{Z_{\mathcal{M}}}, \tag{86}$$

$$Z_{\mathcal{M}} = \int d\vartheta\, \mathcal{L}(d|\vartheta, \mathcal{M})\pi(\vartheta|\mathcal{M}). \tag{87}$$

Model comparison with Kerr requires a Bayes factor and prior-sensitivity analysis, not a comparison of central values.

Acceptance criterion. Gate 11 is passed only after publishing the data selection, forward models, priors, sampler settings, convergence diagnostics, posterior for γ, and evidence ratio. It is formulated mathematically at the equation-and-criterion level.

Toy analyses. Numerical examples based on compressed summary statistics are separated from the physical inference problem and collected in Section 26. No numerical output from those toy constructions is used here as evidence for SSQC or as a substitute for calibrated EHT or gravitational-wave data.

13 Gate 12: predictive power

Parameter fitting and prediction are different operations. A model has predictive power only if parameters constrained by one data set determine the probability distribution of observables that were not used in that

fit. For training data $d_{\rm train}$ and a future or held-out observable $y_{\rm new}$, the posterior predictive distribution is

$$p(y_{\rm new}|d_{\rm train},\mathcal{M}) = \int p(y_{\rm new}|\vartheta,\mathcal{M})p(\vartheta|d_{\rm train},\mathcal{M})\,d\vartheta. \tag{88}$$

For the SSQC model, examples include predicting EHT visibility features from parameters constrained by stellar dynamics, or predicting several ringdown modes after fixing M, a_K, and γ from the inspiral and dominant mode. A separate value of γ must not be selected for every observable unless the model explicitly contains the corresponding environmental dependence.

Out-of-sample performance may be quantified by the log predictive score

$$\mathcal{S}_{\rm pred} = \sum_{k\in{\rm test}} \ln p(d_k|d_{\rm train},\mathcal{M}). \tag{89}$$

The prediction protocol, test set, parameter priors, and tolerated systematic errors must be fixed before examining the held-out measurements.

Acceptance criterion. Gate 12 requires at least one preregistered or genuinely out-of-sample prediction with a calibrated predictive distribution. Agreement obtained after selecting γ from the same observable is parameter estimation, not independent predictive success. This gate is formulated mathematically at the equation-and-criterion level.

14 Gate 13: empirical agreement with observations

Gate 11 defines the inference machinery, while Gate 13 asks whether the completed model actually describes the observations after all nuisance parameters and systematic uncertainties have been included. Agreement must be tested jointly rather than by comparing isolated central values. A posterior predictive check draws replicated data from

$$p(d_{\rm rep}|d_{\rm obs},\mathcal{M}) = \int p(d_{\rm rep}|\vartheta,\mathcal{M})p(\vartheta|d_{\rm obs},\mathcal{M})\,d\vartheta \tag{90}$$

and compares a predefined discrepancy statistic T through

$$p_{\rm PPC} = \Pr[T(d_{\rm rep}) \geq T(d_{\rm obs}) \mid d_{\rm obs},\mathcal{M}]\,. \tag{91}$$

This check must include EHT interferometric data, gravitational-wave strain, and any additional observations claimed to constrain the same parameters. Consistency of geodesic scales alone is insufficient because the observable forward models contain plasma, spin, source dynamics, calibration, and noise.

Agreement is also not model preference. The SSQC evidence must be compared with Kerr and with relevant astrophysical alternatives, including prior sensitivity and injection–recovery tests. A model can be statistically adequate without being preferred, and it can match one observable while failing the joint data set.

Acceptance criterion. Gate 13 requires acceptable posterior predictive diagnostics for every data set in the joint likelihood, no unresolved structured residuals, and a reported evidence comparison with established alternatives. These requirements complete the mathematical formulation of Gate 13 at the equation-and-criterion level.

15 Gate 14: independent replication

Independent replication is an institutional and computational requirement, not another calculation performed by the model's author. A complete reproducibility package must contain the symbolic derivations,

numerical solvers, parameter files, analysis scripts, software versions, random seeds, and cryptographic hashes of all inputs and outputs. For every reported quantity Q_i, an external group should publish an independently generated value $Q_i^{\rm ind}$ and uncertainty.

A normalized replication discrepancy may be defined by

$$D_i = \frac{|Q_i^{\rm ind} - Q_i^{\rm ref}|}{\sqrt{(\sigma_i^{\rm ind})^2 + (\sigma_i^{\rm ref})^2 + \sigma_{i,\rm num}^2}}. \tag{92}$$

Agreement requires both small residuals and agreement on conventions, branches, boundary conditions, and units. Reusing the same unpublished code does not constitute an independent replication.

Acceptance criterion. Gate 14 requires a public report by an unaffiliated researcher or group that rederives the principal analytic formulas and reproduces the numerical tables within declared tolerances. No external confirmation is claimed in this paper. This gate is formulated mathematically at the equation-and-criterion level.

16 Gate 15: data-level Bayesian EHT analysis

Gate 9 specifies the forward radiative-transfer calculation. Gate 15 requires that this calculation be connected to calibrated interferometric data. If $\mathbf{y}$ denotes complex visibilities or closure quantities and $\mathbf{m}(\vartheta)$ denotes the SSQC forward model, a Gaussian approximation has

$$\ln \mathcal{L}_{\rm EHT} = -\frac{1}{2}\left[\mathbf{r}^\dagger \mathbf{C}^{-1}\mathbf{r} + \ln\det(2\pi\mathbf{C})\right], \qquad \mathbf{r} = \mathbf{y} - \mathbf{m}(\vartheta). \tag{93}$$

The actual likelihood may be non-Gaussian and must include station gains, scattering, variability, data selection, and calibration uncertainty. The posterior for γ must be marginalized over the black-hole mass, distance, spin, inclination, electron distribution, magnetic field, and accretion-flow parameters.

Published angular-diameter diagnostic. As a low-dimensional comparison that does not require downloading the full visibility data, use the published M87 emission-ring diameter $d_{\rm EHT} = 42 \pm 3\,\mu$as [4]. This is an emission-ring measurement, not a direct measurement of a vacuum critical curve. For the static metric-null geodesics already derived, define

$$d_{\rm crit} = \frac{2b_c}{D} = \frac{4GM}{c^2 D}\frac{b_c}{R_s}, \qquad \frac{d_{\rm crit}}{R_s/D} = 5.184456 \quad (\gamma = 0.35), \tag{94}$$

where D is the source distance and the angle on the left is in radians. A phenomenological ring-to-critical-curve calibration $\kappa_{\rm ring}$ and a model uncertainty $\sigma_{\rm model}$ permit the diagnostic

$$\chi^2_{\rm diam} = \frac{[\kappa_{\rm ring} d_{\rm crit} - 42\,\mu{\rm as}]^2}{(3\,\mu{\rm as})^2 + \sigma^2_{\rm model}}. \tag{95}$$

All angular quantities in this expression must use the same units. Mass, distance, plasma calibration, and their uncertainties must be stated; $\kappa_{\rm ring} = 1$ is not assumed to be an established transfer result. This comparison is a size diagnostic, not a visibility-level posterior, independent evidence of agreement, or completion of Gate 15.

The calibrated 2017 M87 data are publicly available through the EHT data release [20]. Data availability is therefore not the principal obstruction. For station-based complex gains g_i, a visibility-level forward model must at minimum have the form

$$V_{ij}^{\rm obs}(t,\nu) = g_i(t,\nu) g_j^*(t,\nu) V_{ij}^{\rm SSQC}(\vartheta; t, \nu) + n_{ij}, \tag{96}$$

where V_{ij}^{SSQC} is obtained by ray tracing and polarized transfer, not by rescaling a Kerr radiative-transfer template or inserting a critical impact parameter. The posterior and evidence required for model comparison are

$$p(\vartheta, \mathbf{g}|d, \mathcal{M}_{mSSQC}) \propto \mathcal{L}_{\mathrm{EHT}}(d|\vartheta, \mathbf{g})\pi(\vartheta, \mathbf{g}),$$
$$Z_{\mathrm{SSQC}} = \int d\vartheta \, d\mathbf{g} \, \mathcal{L}_{\mathrm{EHT}}(d|\vartheta, \mathbf{g})\pi(\vartheta, \mathbf{g}). \tag{97}$$

Execution status. The calibrated EHT data release is available, and this article specifies the likelihood and nuisance parameters needed for a visibility-level comparison. The rotating SSQC geometry and associated GRMHD/GRRT simulation library, electron prescription, variability model, gain prior, and injection–recovery validation are not supplied here; accordingly, no calibrated visibility fit or posterior is reported. A scalar ring diameter is only a lowest-dimensional diagnostic.

Acceptance criterion. Gate 15 requires posterior samples from real EHT data, injection–recovery validation, evidence comparison with Kerr-based simulation libraries, and a systematic-error budget. A critical impact parameter or ring diameter does not by itself exhaust this gate. The public data, likelihood equations, nuisance parameters, and acceptance criteria complete the mathematical formulation at the equation-and-criterion level.

17 Gate 16: complete rotating QNM spectrum

Gate 10 states the spectral problem. Gate 16 requires its numerical solution over the astrophysical parameter domain. At least two independent methods—for example continued fractions, spectral collocation, direct integration, or time-domain extraction—must agree. For resolution N, define

$$\epsilon_N^{\ell mn} = \frac{|\omega_{\ell mn}^{(N)} - \omega_{\ell mn}^{(2N)}|}{1 + |\omega_{\ell mn}^{(2N)}|}. \tag{98}$$

The spectrum must include the fundamental mode, overtones, both parity sectors where defined, and the coupled NED degrees of freedom. Isospectrality must not be assumed.

For fields collected into $X^A = (g_{\mu\nu}, A_\mu, \text{additional matter})$, let $\mathcal{E}_A[X] = 0$ denote the complete rotating field equations. A physical QNM operator is the homogeneous linearization

$$\delta\mathcal{E}_A[X_0; \delta X] = \mathcal{D}_A{}^B(\omega, m)\, \delta X_B = 0, \qquad \mathcal{E}_A[X_0] = 0. \tag{99}$$

If the proposed background has a nonzero residual, expansion gives instead

$$\mathcal{D}_A{}^B \delta X_B = -\mathcal{E}_A[X_0] + \mathcal{O}(\delta X^2), \tag{100}$$

which is a forced correction problem, not a QNM eigenvalue problem about a solution. Moreover, a well-defined spectrum requires a hyperbolic principal symbol and boundary conditions compatible with the same matter theory.

Execution status. The physical operator $\mathcal{D}_A{}^B$ is defined by linearizing the coupled slow-rotation equations and the admissible two-invariant matter law. Test-scalar and eikonal frequencies remain useful approximations, whereas the complete SSQC spectrum is the axial–polar–matter spectrum of this operator.

Acceptance criterion. Gate 16 requires independent-code agreement, recovery of the Schwarzschild and Kerr limits, and a converged public frequency table over $(a_K/M, \gamma, q_m/M)$. Eikonal estimates do not count as the full coupled evaluation. The spectral operator, boundary conditions, convergence measure, and comparison limits complete the mathematical formulation of this gate at the equation-and-criterion level.

18 Gate 17: rotating solution with the complete matter field

Gate 5 introduced a metric candidate. Gate 17 requires an exact or controlled perturbative solution of the complete Einstein–NED system. We test the explicit magnetic Kerr–Newman-type field ansatz

$$A_\mu dx^\mu = \frac{q_m \cos\theta}{\Sigma}\left[a_K\, dt - (r^2 + a_K^2)d\phi\right]. \tag{101}$$

At $a_K = 0$, Eq. (101) reduces, up to a gauge-patch sign convention, to the magnetic monopole used in Gate 2. Rotation adds an electric dipole component through A_t and an axisymmetric deformation of A_ϕ.

In addition to the Einstein residual, define the matter-field residual

$$\mathcal{J}^\nu \equiv \nabla_\mu(\mathcal{L}_\mathcal{F} F^{\mu\nu}). \tag{102}$$

For a computational domain $\mathcal{D}$, the normalized joint residual is

$$\mathcal{R}_{\rm rot} = \max_{\mathcal{D}}\left(\frac{\|\mathcal{E}_{\mu\nu}\|}{1+\|G_{\mu\nu}\|} + \frac{\|\mathcal{J}^\nu\|}{1+\|F^{\mu\nu}\|}\right). \tag{103}$$

The electromagnetic invariants, curvature invariants, horizon regularity, axis regularity, and absence of closed timelike curves outside the stated interior region must be checked.

Direct numerical consistency check of the candidate

The check uses $R_s = 1$, $M = 0.5$, core scale $a = 0.35$, magnetic parameter $q_m = 1$, and spin $a_K = 0.20$, corresponding to $a_K/M = 0.4$. The exterior grid covers

$$r \in [r_+ + 0.12, 3], \qquad \theta \in [0.18, \pi - 0.18], \qquad r_+ = 0.9436488. \tag{104}$$

All ten independent symmetric components of $\mathcal{E}_{\mu\nu} = G_{\mu\nu} - 8\pi T_{\mu\nu}$ and all four components of $\mathcal{J}^\nu$ are evaluated by centered finite differences. The same implementation is first tested on the Kerr vacuum and on the exact static SSQC–NED background.

Table 3: Numerical rotating-field consistency check. Residuals are dimensionless in the stated normalization with $R_s = 1$.

Diagnostic	Maximum value
Kerr vacuum normalized Einstein residual	4.29×10^{-7}
Static SSQC normalized Einstein–NED residual	1.35×10^{-6}
Static SSQC NED-equation residual	2.79×10^{-10}
Rotating normalized Einstein–NED residual	1.59×10^{-2}
Rotating NED-equation residual	2.61×10^{-3}
Exterior elementary-flatness error near the axis	2.09×10^{-5}
Minimum sampled $g_{\phi\phi}$	3.76×10^{-2}

The algebraic Kerr-like trial pair does not satisfy the two dynamical tests. Its maximum Einstein–NED residual is approximately 1.2×10^4 times the static validation floor, and its NED residual is approximately 9.4×10^6 times the static NED floor. Changing finite-difference resolution does not remove this order-of-magnitude separation. The tested exterior grid has $g_{\phi\phi} > 0$ and the sampled axis limit satisfies elementary flatness to numerical accuracy. Thus the failure is in the field equations, not in these two exterior regularity diagnostics.

Field-equation rotation formulation. The algebraic Kerr-like metric–field pair is not adopted; the model instead selects the coupled field-equation construction below. The rotating SSQC branch is defined by vanishing order-by-order Einstein–matter residuals, a globally defined field strength, regularity on the axis and horizon, and the slow-rotation functions $h(r)$ and $\omega(r)$. Kerr-like appearance alone is not part of the defining law.

Replacement construction: field-equation slow rotation

The modified Newman–Janis procedure of Azreg-Aïnou is useful for producing regular geometric candidates [12], but it does not remove the requirement to solve the matter field equations. The rejected candidate already has the corresponding Kerr-like mass-function form. Reapplying a different complexification prescription would therefore not address the measured residual.

We instead adopt the generalized Lense–Thirring expansion used for rotating NED solutions [13]. With a bookkeeping parameter ϵ proportional to a_K, write

$$ds^2 = -A_{\rm sc}(r)dt^2 + \frac{dr^2}{A_{\rm sc}(r)} + r^2 d\Omega^2 + 2\epsilon r^2 h(r)\sin^2\theta\, dt\, d\phi + \mathcal{O}(\epsilon^2), \tag{105}$$

$$A_\mu dx^\mu = q_m \cos\theta \left[d\phi - \epsilon \frac{\omega(r)}{r^2} dt \right] + \mathcal{O}(\epsilon^2). \tag{106}$$

The functions $h(r)$ and $\omega(r)$ are independent unknowns. In particular, one must not impose $h = (A_{\rm sc}-1)/r^2$ and $\omega = 1$. For restricted nontrivial NED, that Kerr-like restriction is inconsistent with the field equations already at order ϵ; Maxwell theory is the exceptional case [13].

For comparison with the slow-rotation convention of Ref. [13], introduce rescaled invariants $\mathcal{S} = 2\mathcal{F}$ and $\mathcal{P}_{\rm K} = 2\mathcal{P}$, so that $\mathcal{L} = \mathcal{L}(\mathcal{S}, \mathcal{P}_{\rm K}^2)$, with

$$\mathcal{S} = \frac{1}{2} F_{\mu\nu} F^{\mu\nu}, \qquad \mathcal{P}_{\rm K} = \frac{1}{2} F_{\mu\nu} {}^\star F^{\mu\nu}, \qquad \zeta = r^2 A_{\rm sc}'' - 2A_{\rm sc} + 2. \tag{107}$$

The independent order-ϵ equations reduce to a coupled radial boundary-value problem of the form

$$r^4 A_{\rm sc} h'' + 4r^3 A_{\rm sc} h' - r^2 \zeta h - \zeta \omega = 0, \tag{108}$$

$$\mathfrak{A}(r)\omega'' + \mathfrak{B}(r)\omega' + \mathfrak{C}(r)\omega - 2r^6 \mathcal{L}_{\mathcal{S}} h = 0, \tag{109}$$

where the constitutive coefficients $\mathfrak{A}$, $\mathfrak{B}$, and $\mathfrak{C}$ are given explicitly in Appendix B. Their evaluation requires the static-background value of $\mathcal{L}_{\mathcal{S}}$ and the independent second-invariant response $\mathcal{L}_{\mathcal{P}_{\rm K}^2}$; the latter is not fixed by the purely magnetic static solution alone. Regularity at the outer horizon, regular gauge patches on the axis, and asymptotic conditions including $h = -2M/r^3 + o(r^{-3})$ provide the boundary data.

Symbolic asymptotic analysis. Direct differentiation of the exact static lapse gives

$$\zeta(r) = \frac{42Ma^4 r^6}{(r^4+a^4)^{11/4}} = \frac{42Ma^4}{r^5} + \mathcal{O}(r^{-9}). \tag{110}$$

Define the asymptotic coefficients by

$$h(r) = \frac{1}{r^3}\sum_{j=0}^{\infty} \frac{h_j}{r^j}, \qquad \omega(r) = \sum_{j=0}^{\infty} \frac{w_j}{r^j}. \tag{111}$$

Substitution into Eq. (108) yields at the first two nontrivial orders

$$\frac{4h_1}{r^2} + \frac{10h_2 - 8Mh_1}{r^3} + \mathcal{O}(r^{-4}) = 0, \qquad h_1 = h_2 = 0, \quad h_0 = -2M. \tag{112}$$

The last equality fixes the spin normalization through the boundary law $h \sim -2M/r^3$. These metric recursions follow from the displayed equation and do not require identifying the rotating ansatz with Kerr.

A separate submitted symbolic calculation reports $w_0 = 1$, $w_1 = 0$, and $w_2 = -(7\,2^{1/4}/16)h_0$. These electromagnetic relations cannot yet be assigned to the exact SSQC system: although the coefficient expressions are given in Appendix B, a two-invariant constitutive law (including $\mathcal{L}_{\mathcal{P}_{\rm K}^2}$), the units, and

reproducible solution data are still needed. Here $\omega \to 1$ is retained only as a proposed asymptotic boundary choice to test against that system, not as a derived SSQC matter solution.

Likewise, the positive symbolic expression

$$\mathcal{H}_{\rm pow} = \frac{35}{8}\mathcal{F}^{3/4} > 0 \quad \text{for} \quad \mathcal{L}_{\rm pow} = \mathcal{F}^{7/4}, \quad \mathcal{F} > 0 \tag{113}$$

belongs to a unit-normalized power-law constitutive test. It is not the exact reconstructed Lagrangian (14); its positivity cannot replace Eq. (51) or cure the SSQC core failure.

For fixed $r > 0$, taking $a \to 0$ gives the Schwarzschild lapse and $\zeta \to 0$. With $h = -2M/r^3$, the metric sector then reproduces Kerr to first order in spin. This geometric limit is not a proof that the charged NED field has a regular Maxwell or Kerr–Newman limit: at fixed finite $\mathcal{F}$ and q_m, the reconstructed $\mathcal{L}$ tends to zero.

This is the replacement rotating problem formulated by the article. It is substantially more constrained than the algebraic substitution $M \mapsto m(r)$: the order-ϵ Einstein and NED residuals vanish as functions of r when the coupled boundary-value equations are imposed. Equations (108)–(109), together with the explicit coefficients in Appendix B, rather than a Kerr-like mass substitution, define the proposed slow-rotation boundary-value problem. Completing it still requires a specified two-invariant matter Lagrangian and a solution satisfying all boundary conditions. On the subextremal black-hole branch, the same framework may be studied as an exterior problem because the minimal-source constitutive sign change lies inside r_+.

18.1 Horizon shielding and the domain of the slow-rotation problem

Two logically distinct questions must be separated. An event horizon may causally shield an exterior initial-boundary-value problem from a defective central region, but it cannot make an incorrect metric–matter pair satisfy the field equations and it cannot restore global hyperbolicity of the full spacetime. The direct numerical check above was performed for $\gamma = 0.35$, which is on the two-horizon black-hole branch, and on a grid entirely outside r_+. Its nonzero residual therefore disproves the claim that the previously tested Kerr-like ansatz becomes a solution merely because an outer horizon is present.

The more limited shielding statement can be made precisely. In the static geometry,

$$r_{\rm hyp} = \left(\frac{6}{5}\right)^{1/4}\gamma R_s, \qquad r_{\rm max} = 2^{1/4}\gamma R_s, \tag{114}$$

where $r_{\rm max}$ is the position of the maximum that generates the two horizon roots. On the subextremal branch the outer root satisfies $r_+ > r_{\rm max}$, while

$$\frac{r_{\rm hyp}}{r_{\rm max}} = \left(\frac{3}{5}\right)^{1/4} \simeq 0.8801. \tag{115}$$

Consequently,

$$0 < \gamma < \gamma_{\rm H} \quad \Longrightarrow \quad r_{\rm hyp} < r_{\rm max} < r_+. \tag{116}$$

The displayed constitutive sign failure is therefore hidden behind the outer horizon for every static subextremal black hole. In the first-order slow-rotation metric (105), the horizon radius is unchanged at order ϵ because the diagonal static sector changes only at order ϵ^2.

This ordering permits a *conditional exterior effective problem* on

$$\mathcal{D}_{\rm ext} = \{r \geq r_+\}, \tag{117}$$

provided that the new slow-rotation solution satisfies the Einstein–matter equations and every characteristic condition throughout $\mathcal{D}_{\rm ext}$. It does not validate the interior source, collapse, maximal extension, Cauchy-horizon evolution, or global stability.

The branch-dependent status is therefore:

1. For $0 < \gamma < \gamma_{\rm H}$, the slow-rotation boundary-value problem defines the conditions that a candidate exterior rotating solution must satisfy, including regular ingoing data at r_+ and a vanishing order-ϵ residual.

2. At $\gamma = \gamma_{\rm H}$, the horizon is degenerate. The near-horizon scaling and the compatibility of slow rotation with extremality require a separate expansion; the subextremal argument cannot simply be copied.

3. For $\gamma > \gamma_{\rm H}$, no horizon shields the region $r < r_{\rm hyp}$. The present minimal NED then fails as a global matter completion, and a beyond-minimal-NED action or additional degree of freedom is mandatory before a causal evolution can be defined.

Accordingly, the present minimal completion can at most be used as a conditional exterior model of the black-hole branch. EHT ray tracing and black-hole QNM boundary conditions may be considered only after the slow-rotation exterior equations have been solved. Such calculations would test exterior phenomenology; they would not prove that the regular interior is dynamically realizable. Horizonless compact objects are outside the current domain of applicability.

19 Gate 18: complete linear-stability assessment

Gate 6 gives the coupled perturbation equations. Gate 18 requires the spectral and time-domain analysis. Define the spectral growth rate

$$\alpha_{\rm spec} = \sup_{\ell,n,\pm} \operatorname{Im} \omega_{\ell n}^{(\pm)}. \tag{118}$$

Mode stability requires $\alpha_{\rm spec} \leq 0$ for the complete admissible spectrum. This condition must be supplemented by hyperbolicity, absence of ghost and gradient instabilities, and bounded time-domain evolution for compact initial data. Near-extremal horizons and any Cauchy horizon require separate treatment.

Stability law. The complete stability rule is defined as follows: axial, polar, and matter sectors are governed by the enlarged two-invariant source, with a hyperbolic principal symbol, regular horizon data, nonpositive mode growth, and bounded time-domain evolution. Equations (51)– (52) identify why the minimal one-invariant representation is not the physical global branch. The SSQC stability law therefore combines the regular geometry with the admissible $\mathcal{L}(\mathcal{F}, \mathcal{P}^2)$ matter sector rather than tuning the excluded minimal Lagrangian.

20 Gate 19: quantitative predictions for future observations

Gate 12 defines predictive power in general. Gate 19 requires a dated, numerical prediction for observations not yet included in the fit. The model must publish a vector such as

$$\mathbf{Y}_{\rm future} = (d_{\rm ring}, A_{\rm asym}, f_{220}, \tau_{220}, f_{330}, \tau_{330}, \Delta t_{\rm delay}, \ldots) \tag{119}$$

with a joint covariance and a specified observing configuration. Suitable targets may include a future EHT or ngEHT campaign, the Einstein Telescope, Cosmic Explorer, or LISA. Predictions must be conditioned on a posterior obtained without the future data.

Acceptance criterion. Gate 19 requires a timestamped prediction, frozen analysis choices, and a subsequent blind comparison. Changing the parameter range after seeing the new data invalidates the predictive test. The prediction vector, timestamp rule, frozen-analysis condition, and blind-comparison criterion complete the mathematical formulation of Gate 19.

21 Gate 20: comparison with alternative regular metrics

The SSQC model must be compared with established regular metrics using the same data, priors, and nuisance models. Representative examples include the Bardeen, Hayward, and Dymnikova backgrounds [9, 10, 11]. Their mass functions are

$$m_{\mathrm{B}}(r) = \frac{Mr^3}{(r^2+g^2)^{3/2}}, \tag{120}$$

$$m_{\mathrm{H}}(r) = \frac{Mr^3}{r^3+2M\ell^2}, \tag{121}$$

$$m_{\mathrm{D}}(r) = M\left[1-e^{-r^3/r_0^3}\right], \tag{122}$$

$$m_{\mathrm{SSQC}}(r) = \frac{Mr^3}{(r^4+a^4)^{3/4}}. \tag{123}$$

Table 4: Structural comparison before observational fitting.

Model	Central behavior	Leading exterior correction	Source status
Bardeen	de Sitter-like	power-law, starting beyond $1/r$	magnetic NED interpretation available
Hayward	de Sitter-like	power-law	usually phenomenological effective source
Dymnikova	de Sitter-like	exponentially suppressed	vacuum-like anisotropic source
SSQC	de Sitter-like	$\mathcal{O}(r^{-5})$ in $A(r)$	reconstructed magnetic NED without Maxwell weak-field limit

"Better" must be defined by predictive accuracy, evidence, parameter economy, stability, formation, and theoretical consistency. A smaller residual after adding more nuisance parameters is not sufficient.

Acceptance criterion. Gate 20 requires a common benchmark pipeline and published Bayes factors or out-of-sample scores for Kerr, Bardeen, Hayward, Dymnikova, and SSQC models. No superiority claim is made here. This gate is formulated mathematically at the equation-and-criterion level.

22 Gate 21: data-level Bayesian gravitational-wave analysis

Gate 16 must first supply a rotating waveform family. For detector k with one-sided noise spectrum $S_{n,k}(f)$, the standard frequency-domain likelihood used in gravitational-wave analyses such as GW150914 [21] contains the noise-weighted inner product

$$\ln \mathcal{L}_{\mathrm{GW}} = -\frac{1}{2}\sum_k (d_k - h_k | d_k - h_k)_k + \mathrm{const}, \tag{124}$$

$$(a|b)_k = 4\,\mathrm{Re}\int_{f_{\min}}^{f_{\max}} \frac{\tilde{a}(f)\tilde{b}^*(f)}{S_{n,k}(f)}\,df. \tag{125}$$

The measured strain is modeled as $d_k(t) = h_k(t;\vartheta) + n_k(t)$, with

$$h_k(t;\vartheta) = F_k^+(\alpha,\delta,\psi)h_+(t-\tau_k;\vartheta) + F_k^\times(\alpha,\delta,\psi)h_\times(t-\tau_k;\vartheta), \tag{126}$$

where $F_k^{+,\times}$ are detector antenna responses and τ_k is the arrival-time delay. With stationary Gaussian noise and a specified PSD, Eqs. (124)–(125) define the strain likelihood. The associated posterior and evidence are

$$p(\vartheta|d,\mathcal{M}) = \frac{\mathcal{L}_{\rm GW}(d|\vartheta,\mathcal{M})\pi(\vartheta|\mathcal{M})}{Z_{\mathcal{M}}}, \qquad Z_{\mathcal{M}} = \int d\vartheta\, \mathcal{L}_{\rm GW}\pi. \tag{127}$$

This is an analysis prescription; no calibrated SSQC waveform or measured strain likelihood is evaluated here. A damped test signal or a photon-orbit frequency is not a substitute for $h_{+,\times}$.

The analysis must marginalize over calibration, sky position, distance, orientation, masses, spins, merger time, phase, noise nonstationarity, and the ringdown start time. The prior on γ must be declared before the fit.

Acceptance criterion. Gate 21 requires analysis of public strain data with injection campaigns, sampler convergence, waveform-systematic estimates, a posterior for γ, and evidence relative to general-relativistic waveforms. Agreement of approximate frequencies is not a LIGO test. This gate is formulated mathematically at the equation-and-criterion level.

23 Gate 22: dynamical formation by collapse

A static solution is not an astrophysical formation model. Consider a time-dependent spherical geometry

$$ds^2 = -e^{2\Phi(t,r)}F(t,r)dt^2 + \frac{dr^2}{F(t,r)} + r^2 d\Omega^2, \qquad F(t,r) = 1 - \frac{2m(t,r)}{r}. \tag{128}$$

The Einstein–matter evolution must start from regular initial data that represent an admissible collapsing configuration and approach the SSQC mass function, if it does so at all:

$$\lim_{t\to t_f} m(t,r) = \frac{Mr^3}{(r^4+a^4)^{3/4}}. \tag{129}$$

The origin of the magnetic charge and of the effective NED state must be explained. Imposing the final exotic stress tensor by hand at late time is not a formation derivation.

Acceptance criterion. Gate 22 requires a well-posed initial-value problem, constraint-satisfying initial data, numerical collapse without curvature divergence, formation of the claimed horizon structure, and robustness under perturbations of the initial data. This gate is formulated mathematically at the equation-and-criterion level.

24 Gate 23: semiclassical and quantum corrections

Classical regularity does not demonstrate quantum consistency. A first semiclassical analysis requires a specified quantum state and a renormalization prescription [23, 24]. A first diagnostic compares curvature with the Planck scale,

$$\epsilon_{\rm Q}(r) = \ell_{\rm P}^4 K(r), \qquad \epsilon_{\rm Q}(0) = \frac{24\ell_{\rm P}^4 R_s^2}{a^6}. \tag{130}$$

When $\epsilon_{\rm Q}$ is not small, higher-curvature operators cannot be discarded solely on dimensional grounds. The semiclassical field equation is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} + \sum_i c_i \ell_{\rm P}^{2n_i}\mathcal{H}_{\mu\nu}^{(i)} = 8\pi G\left(T_{\mu\nu}^{\rm NED} + \langle T_{\mu\nu}\rangle_{\rm ren}\right). \tag{131}$$

The state dependence of $\langle T_{\mu\nu}\rangle_{\rm ren}$, trace anomaly, Hawking flux, and backreaction on both horizons must be evaluated.

For $a = \gamma R_s$,

$$\epsilon_{\rm Q}(0) = \frac{24}{\gamma^6}\left(\frac{\ell_{\rm P}}{R_s}\right)^4. \tag{132}$$

At $\gamma = 0.35$, this dimensionless curvature estimate gives approximately 1.17×10^{-149} for $M = M_\odot$, 7.93×10^{-157} for $M = 62M_\odot$, and 6.56×10^{-189} for $M = 6.5 \times 10^9 M_\odot$. Thus local higher-curvature corrections controlled solely by $\ell_{\rm P}^4 K$ are dimensionally negligible for these astrophysical masses.

Hawking evaporation and a prescribed core-scale law. As a kinematic scaling exercise, suppose the core scale is postulated to vary with the instantaneous mass according to

$$a(M) = a_0\left(\frac{M}{M_0}\right)^p, \qquad R_s(M) = R_{s0}\frac{M}{M_0}, \qquad \gamma(M) = \gamma_0\mu^{p-1}, \quad \mu = \frac{M}{M_0}, \quad \gamma_0 = \frac{a_0}{R_{s0}}. \tag{133}$$

On the subextremal black-hole branch, let $x_+(\gamma) = r_+/R_s$ be the largest positive root of Eq. (27). The ratio in question is then

$$\frac{r_{\rm hyp}}{r_+} = \frac{(6/5)^{1/4}\gamma(M)}{x_+(\gamma(M))}, \qquad \gamma(M) < \gamma_{\rm H}. \tag{134}$$

For $p > 1$, this ratio tends to zero as $\mu \to 0^+$ along the subextremal static family, because $\gamma \to 0$ and $x_+(\gamma) \to 1$. For $p = 1$, $\gamma = \gamma_0$ and the ratio is constant while a horizon exists (for $\gamma_0 = 0.35$, it is approximately 0.3706). For $p < 1$, γ increases as the mass decreases; if $\gamma_0 < \gamma_{\rm H}$, the family reaches $\gamma = \gamma_{\rm H}$ at

$$\mu_{\rm H} = \left(\frac{\gamma_{\rm H}}{\gamma_0}\right)^{1/(p-1)}, \tag{135}$$

after which the static metric has no horizon. For example, with $\gamma_0 = 0.35$ and $p = 1/2$, $\mu_{\rm H} \simeq 0.318$.

These limits describe only geometric shielding in a prescribed family of static metrics; they do not show that the constitutive law becomes hyperbolic at any finite $M > 0$. Indeed, for every $a(M) > 0$ the exact one-invariant SSQC source still has $\mathcal{H} < 0$ for $0 < r < r_{\rm hyp}$. At $M \to 0$, both the horizon and core scale vanish, so a zero ratio is not by itself restoration of global hyperbolicity of a nonzero-mass black hole. A self-consistent evaporation model must derive a(M) dynamically and include the changing matter couplings, Hawking flux, and backreaction in Eq. (131); the scaling ansatz above does not do so.

Likewise, this scaling argument does not solve the slow-rotation boundary-value problem. The coefficients and reproducible solution data needed to verify the reported $\omega(r)$ residual are not specified here; Gate 5 therefore remains unsolved, and no rotating evaporation solution or residual at the quoted level is claimed.

Acceptance criterion. Gate 23 requires either a controlled regime in which all quantum corrections are demonstrably small or a self-consistent corrected solution. Merely having finite classical curvature does not pass this gate. The renormalized stress tensor and backreaction problem are specified by Eq. (131) and its acceptance criterion. Gate 23 is therefore formulated, not completed.

25 Scope and reproducibility

The present work establishes the static background results and formulates the remaining gates. The official calibrated EHT release and public GW150914 strain data are existing observational data products [20, 22]; this article identifies them as such, but does not report a data-level likelihood or posterior based on them. The remaining gates also specify the requirements for the rotating boundary-value problem, coupled QNM spectrum, collapse initial data, and semiclassical stress tensor.

25.1 Numerical benchmarks

For the SSQC metric at $\gamma = 0.35$, direct bracketed root finding gives

$$\frac{r_-}{R_s} = 0.2182994602, \qquad \frac{r_+}{R_s} = 0.9883659438. \tag{136}$$

The outer surface gravity and temperature are

$$\kappa_+ R_s = \frac{1}{2}|A'_{\rm sc}(r_+)|R_s = 0.4823892657, \qquad T_H R_s = \frac{\kappa_+ R_s}{2\pi} = 0.0767746361. \tag{137}$$

The outer circular null orbit and critical impact parameter of the SSQC metric are

$$\frac{r_{\rm ph}}{R_s} = 1.4920855318, \qquad \frac{b_c}{R_s} = 2.5922280751. \tag{138}$$

25.1.1 Extended benchmark at the reference parameter point

Use the dimensionless normalization

$$R_s = 1, \qquad \gamma = 0.35, \qquad a = 0.35, \qquad M = 0.5, \qquad q_m = 1. \tag{139}$$

The horizon and outer photon-orbit values in Eqs. (136) and (138) are supplemented by the inner circular null orbit

$$r_{\rm ph}^{(\rm in)}/R_s = 0.3210004053, \qquad A_{\rm sc}(r_{\rm ph}^{(\rm in)}) = -0.6089002848. \tag{140}$$

Because this orbit lies in the nonstatic region between the two horizons, $A_{\rm sc} < 0$ and the usual real impact parameter $b_c = r_{\rm ph}/\sqrt{A_{\rm sc}(r_{\rm ph})}$ is not defined there. The same parameter point gives

$$r_{\rm hyp}/R_s = 0.3663222988, \qquad R(0)R_s^2 = 279.8833819242, \qquad K(0)R_s^4 = 13055.7845795544, \tag{141}$$

in agreement with Eqs. (53), (23), and (24).

The absence of a Maxwell weak-field limit can also be seen numerically. For the parameters in Eq. (139), Table 5 shows that $\mathcal{L}/\mathcal{F} \to 0$, whereas $\mathcal{L}/\mathcal{F}^{7/4}$ approaches 6.0249863×10^{-3}.

Table 5: Weak-field benchmark for the reconstructed SSQC Lagrangian in the normalization with $R_s = 1$.

$\mathcal{F}$	$\mathcal{L}$	$\mathcal{L}/\mathcal{F}$	$\mathcal{L}/\mathcal{F}^{7/4}$
10^{-1}	1.065807×10^{-4}	1.065807×10^{-3}	5.993472×10^{-3}
10^{-2}	1.904268×10^{-6}	1.904268×10^{-4}	6.021823×10^{-3}
10^{-3}	3.387921×10^{-8}	3.387921×10^{-5}	6.024670×10^{-3}
10^{-4}	6.024955×10^{-10}	6.024955×10^{-6}	6.024955×10^{-3}
10^{-5}	1.071410×10^{-11}	1.071410×10^{-6}	6.024983×10^{-3}
10^{-6}	1.905268×10^{-13}	1.905268×10^{-7}	6.024986×10^{-3}

For a separate kinematical dyonic check, take $q_e = 0.5$ and assume the Coulomb profiles $E = q_e/r^2$ and $B = q_m/r^2$. With the convention in Eq. (43),

$$\mathcal{F} = \frac{q_m^2 - q_e^2}{2r^4}, \qquad \mathcal{P} = -\frac{q_e q_m}{r^4}, \qquad \mathcal{X} = \mathcal{P}^2. \tag{142}$$

The corresponding values are listed in Table 6. The factor in $\mathcal{P}$ is fixed by the definition $\mathcal{P} = F_{\mu\nu}{}^\star F^{\mu\nu}/4$; using $-q_e q_m/(2r^4)$ would be inconsistent with that convention.

An exploratory scan of simple corrections of the form $\mathcal{L}_{\rm base} + \alpha \mathcal{X}^2 f(\mathcal{F})$ did not identify an admissible candidate: sampled choices violated positivity of $\mathcal{L}_\mathcal{F}$ or failed the adopted characteristic tests. This negative scan is only diagnostic because the function family, parameter ranges, resolution, and complete

Table 6: Kinematical two-invariant benchmark for $q_m = 1$ and $q_e = 0.5$. This table does not assert a dyonic SSQC field-equation solution.

r/R_s	$\mathcal{F}R_s^2$	$\mathcal{P}R_s^2$	$\mathcal{X}R_s^4$
0.1000	3.750000×10^{3}	-5.000000×10^{3}	2.500000×10^{7}
0.2000	2.343750×10^{2}	-3.125000×10^{2}	9.765625×10^{4}
0.3663	2.082466×10^{1}	-2.776621×10^{1}	7.709623×10^{2}
0.5000	6.000000	-8.000000	6.400000×10^{1}
1.0000	3.750000×10^{-1}	-5.000000×10^{-1}	2.500000×10^{-1}
2.0000	2.343750×10^{-2}	-3.125000×10^{-2}	9.765625×10^{-4}
5.0000	6.000000×10^{-4}	-8.000000×10^{-4}	6.400000×10^{-7}

principal-symbol eigenvalues have not been archived. In particular, the sign of the Hessian determinant alone is not a sufficient hyperbolicity criterion. No two-invariant completion is therefore claimed.

A trial integration of the slow-rotation radial system was also carried from $r/R_s = 0.9984$ to 10, returning

$$h(10R_s) = 2.672783 \times 10^{-1}, \qquad \omega(10R_s) = 1.300095. \tag{143}$$

This confirms that the numerical integration infrastructure runs outside the outer horizon, but it does not solve the boundary-value problem. The required asymptotic condition gives $h(10R_s) \simeq -2M/(10R_s)^3 = -10^{-3}R_s^{-2}$, which differs in sign and magnitude from the trial endpoint. A validated slow-rotation solution still requires the missing constitutive coefficients, horizon data, asymptotic shooting or collocation, and residual convergence.

25.2 Scope of reported results

The exact static background, effective magnetic source, local regularity, and associated numerical checks are reported in this article. The calibrated 2017 M87 EHT data release is an existing public observational data product; the article specifies a visibility-level comparison but does not report an EHT likelihood or posterior. The rotating boundary-value problem, complete coupled stability analysis, gravitational-wave inference, collapse, and semiclassical backreaction are represented by the stated equations and acceptance criteria, rather than by completed calculations. The verification matrix below summarizes the scope of these results.

26 Controlled toy demonstrations and diagnostics

Toy, not physical. This section collects all deliberately simplified numerical examples. Their scope is fixed by their inputs: a compressed summary vector is not a joint data-level likelihood, generated visibilities are not calibrated EHT data, an eikonal formula is not a full coupled QNM eigenproblem, and a generated damped sinusoid is not detector strain. The calculations may therefore be used as pipeline demonstrations, but they do not change the validation status assigned in the preceding status assessment.

26.1 Gate 11: compressed-summary posterior demonstration

Toy, not physical. Consider a symbolic compressed data vector $d_{\rm toy}$ containing a ring size, a ringdown frequency, and a damping time, with diagonal Gaussian expression

$$\ln \mathcal{L}_{\rm toy} = -\frac{1}{2}\sum_i \frac{[d_i - m_i(\vartheta)]^2}{\sigma_i^2}. \tag{144}$$

Equation (144) becomes a physical likelihood only after every $m_i(\vartheta)$ has been defined by the same validated forward model. Here the EHT entry is a bright-ring summary rather than calibrated visibility

data, while the two ringdown entries have no specified strain data set, waveform, covariance, or detector-noise model.

Toy priors and numerical posterior outputs are intentionally not reported, because they are generator-dependent and could be mistaken for observational constraints. Likewise, writing

$$B_{\mathrm{SSQC/Kerr}} = \frac{Z_{\mathrm{SSQC}}}{Z_{\mathrm{Kerr}}}, \qquad Z_{\mathcal{M}} = \int \mathcal{L}(d|\vartheta, \mathcal{M})\pi(\vartheta|\mathcal{M})d\vartheta \tag{145}$$

does not produce a physical Bayes factor without both evidence calculations, their numerical errors, and prior-sensitivity tests. This construction is retained only as a methodological demonstration and is not observational evidence for SSQC.

26.2 Gate 15: generated visibility-level Bayesian demonstration

Toy, not physical. The illustrative test uses controlled generated visibility samples with independent Gaussian noise σ_V. Introduce the dimensionless baseline $\widehat{b} = b/(1\,\mathrm{G}\lambda)$ and define

$$V_{\mathrm{toy}}(\widehat{b};\gamma) = \exp\left[-\frac{1}{2}(\widehat{b}\, r_{\mathrm{ring}}(\gamma))^2\right]\cos\left[2\pi\widehat{b}\, r_{\mathrm{ring}}(\gamma)\right], \tag{146}$$

$$r_{\mathrm{ring}}(\gamma) = b_c^{\mathrm{toy}}(\gamma)\theta_g, \qquad b_c^{\mathrm{toy}}(\gamma) = b_{c,0}\left[1 + \epsilon_{\mathrm{toy}}\frac{\gamma-\gamma_0}{\gamma_0}\right]. \tag{147}$$

For generated values $V_{\mathrm{obs},i}$, the adopted likelihood is

$$\mathcal{L}_{\mathrm{syn}}(\gamma) = \prod_{i=1}^{N_{\mathrm{toy}}} \frac{1}{\sqrt{2\pi\sigma_V^2}}\exp\left[-\frac{[V_{\mathrm{obs},i} - V_{\mathrm{toy}}(\widehat{b}_i;\gamma)]^2}{2\sigma_V^2}\right]. \tag{148}$$

Numerical posterior and evidence outputs from this generator are deliberately omitted. Any such values would quantify only recovery inside the selected toy generator and noise model; they would not be a Bayes factor from the 2017 EHT visibility data and would not close Gate 15.

26.3 Gate 16: illustrative rotating eikonal catalogue

Diagnostic approximation, not a physical SSQC spectrum. The illustrative prescription combines the static photon-orbit diagnostic [18] with an imposed rotational frequency shift:

$$\omega_{\ell mn}^{\mathrm{eik}} = \left(\ell + \frac{1}{2}\right)\Omega_{\mathrm{ph}} + m\Omega_{\mathrm{rot}} - i\left(n + \frac{1}{2}\right)\lambda_{\mathrm{ph}}, \tag{149}$$

with

$$\Omega_{\mathrm{ph}} = \frac{\sqrt{A(r_{\mathrm{ph}})}}{r_{\mathrm{ph}}}, \quad \lambda_{\mathrm{ph}} = \sqrt{\frac{A(r_{\mathrm{ph}})[2A(r_{\mathrm{ph}}) - r_{\mathrm{ph}}^2 A''(r_{\mathrm{ph}})]}{2r_{\mathrm{ph}}^2}}, \quad \Omega_{\mathrm{rot}} = \frac{2a_K M}{r_{\mathrm{ph}}^3}. \tag{150}$$

Here primes denote derivatives with respect to the areal radius r. Differentiating Eq. (3) twice gives

$$A_{\mathrm{sc}}''(r) = -R_s\frac{2a^8 - 17a^4r^4 + 2r^8}{(r^4+a^4)^{11/4}}. \tag{151}$$

At $\gamma = 0.35$, solving $rA_{\mathrm{sc}}' - 2A_{\mathrm{sc}} = 0$ for the outer null orbit and evaluating these expressions yields

$$r_{\mathrm{ph}}/R_s = 1.4920855318, \qquad A_{\mathrm{sc}}(r_{\mathrm{ph}}) = 0.3313149389, \tag{152}$$

$$R_s^2 A_{\mathrm{sc}}''(r_{\mathrm{ph}}) = -0.5817277954, \qquad R_s\Omega_{\mathrm{ph}} = 0.3857685246, \tag{153}$$

$$R_s\lambda_{\mathrm{ph}} = 0.3816712816. \tag{154}$$

Thus the quoted second derivative follows directly from the SSQC lapse; it is not a fitted input or a derivative with respect to r/M.

To convert the geometrized frequencies to physical units, define $\widehat{\omega} = R_s\omega$ and $t_s = R_s/c = 2GM/c^3$. Then

$$f = \frac{\mathrm{Re}\,\widehat{\omega}}{2\pi t_s}, \qquad \tau = \frac{t_s}{|\,\mathrm{Im}\,\widehat{\omega}|}, \qquad t_s \simeq 6.107608774 \times 10^{-4}\,\mathrm{s} \quad (M = 62M_\odot), \tag{155}$$

using $GM_\odot/c^3 \simeq 4.925490947\,\mu$s. For $\chi = a_K/M$, the imposed shift is $R_s\Omega_{\rm rot} = \chi/[2(r_{\rm ph}/R_s)^3]$. The recomputed illustrative catalogue is shown in Table 7.

Table 7: Recomputed photon-orbit diagnostics for $\gamma = 0.35$ and $M = 62M_\odot$. Nonzero-spin rows use an imposed linear shift, not a solved rotating SSQC spacetime.

ℓ	m	n	$\chi = a_K/M$	f (Hz)	τ (ms)
2	2	0	0.0	251.3	3.20
2	2	0	0.4	282.7	3.20
3	3	0	0.4	398.9	3.20
4	4	0	0.4	515.1	3.20

Equation (149) is a photon-orbit approximation. It does not contain the axial, polar, or NED perturbation operators and cannot produce the complete rotating spectrum requested by Gate 16. In particular, the static orbit and Lyapunov exponent are held fixed in the nonzero-spin rows, which are not established as controlled slow-rotation predictions. The low multipoles shown also carry no demonstrated eikonal error bound. The table is therefore retained as an illustrative diagnostic only.

26.4 Gate 21: generated damped-sinusoid recovery

Toy, not physical. The illustrative waveform family is

$$h_{\rm syn}(t;\gamma) = \Theta(t-t_0)\exp\left[-\frac{t-t_0}{\tau(\gamma)}\right]\sin[2\pi f(\gamma)(t-t_0)]\,, \tag{156}$$

where

$$f(\gamma) = f_0\left[1+\alpha_f\frac{\gamma-\gamma_0}{\gamma_0}\right], \qquad \tau(\gamma) = \tau_0\left[1+\alpha_\tau\frac{\gamma-\gamma_0}{\gamma_0}\right]. \tag{157}$$

Numerical recovery values are deliberately omitted because they depend only on the selected generator and may be mistaken for detector results. No H1/L1 strain likelihood, detector PSD, calibration model, start-time marginalization, or evidence integral appears in Eqs. (156)– (157). Gate 21 is formulated mathematically through the strain likelihood, nuisance parameters, and evidence criterion.

Table 8: Scope of the four illustrative toy demonstrations and diagnostics.

Gate	Demonstrated result	Validation status
11	Posterior demonstration from three compressed summary statistics	Toy, not physical; no common validated forward model or data-level likelihood.
15	Parameter recovery from controlled generated visibility samples	Toy, not physical; full EHT gate formulated mathematically.
16	Eikonal catalogue over selected ℓ, m, n, a_K values	Photon-orbit diagnostic; full rotating QNM gate formulated mathematically.
21	Generated damped-sinusoid pipeline	Toy only; no strain-data inference.

27 Dependency structure of the validation framework

The remaining gates are not independent checkboxes. Several cannot be evaluated independently of the field-equation construction:

1. Gates 10, 16, and 21 use the rotating metric–field law. The Kerr-like trial geometry is replaced by the coupled slow-rotation boundary-value problem on $r \geq r_+$. Its residual and boundary conditions define the input required by the QNM and gravitational-wave calculations.

2. Gates 9 and 15 require a valid rotating geometry plus a plasma and instrument model. The slow-rotation construction must first be solved and then extended to the spin range relevant to M87*. A spherical critical curve cannot replace GRMHD/GRRT radiative-transfer calculations and interferometric sampling.

3. Gates 6 and 18 use a hyperbolic matter system. The reconstructed one-invariant NED has a strong-field sign change behind the outer horizon. The two-invariant constitutive class supplies the framework for the global and horizonless stability analysis, while the subextremal exterior problem is posed directly on the black-hole branch.

4. Gates 12 and 19 require a completed forward model and a posterior fixed without the test data. Neither prerequisite currently exists.

5. Gate 14 cannot be completed by additional calculations from the same author; independence is part of its definition.

6. Gate 13 requires the completed EHT and gravitational-wave likelihoods, and Gate 22 requires a new time-dependent collapse calculation.

This hierarchy separates the exact static results, the constructed field-equation laws, and their observational applications. It prevents a geometric diagnostic from being confused with a complete data-level test.

28 Verification matrix

No.	Gate	Current status	Required next result
1	Action	Derived, static	Independent symbolic substitution and normalization check.
2	Source	Derived, effective NED	Causality, weak-field completion, and formation model.
3	Geodesic completeness	Local extension derived	Maximal global atlas and continuation across all horizons.
4	Central smoothness	Derived locally	Independent Cartesian regularity check.
5	Rotation	Metric recursions derived; BVP unsolved	$h(r)$ and $\omega(r)$ obey horizon and asymptotic boundary laws with vanishing order-by-order residual.
6	Stability	Exterior constitutive signs established; full stability open	Coupled spectrum and characteristic-cone analysis; admissible global matter completion.
7	Delayed wave response	Formulated mathematically	Reflection coefficient, delay, and transfer-function criteria specified.
8	Global causal structure	Formulated mathematically	Conformal atlas and Cauchy-horizon criteria specified.
9	EHT	Forward problem formulated	Rotating geometry, GRMHD/GRRT map, visibilities, gains, and likelihood specified.

No.	Gate	Current status	Required next result
10	LIGO/QNM	Defined on the rotating exterior framework	Apply the coupled perturbation operator to the selected observational parameter set.
11	Bayesian analysis	Data-level protocol specified	Define the physical forward models, data products, covariance, chains, posterior, and evidence.
12	Predictive power	Formulated mathematically	Preregistered or held-out prediction with a calibrated predictive distribution.
13	Observational agreement	Formulated mathematically	Joint posterior predictive checks, systematic-error assessment, and comparison with Kerr.
14	Independent replication	Formulated mathematically	Independent analytic derivation and numerical reproduction criterion.
15	Full Bayesian EHT analysis	Official calibrated data identified	Visibility-level posterior using a validated rotating GRRT model, with gain marginalization, injection–recovery tests, and evidence comparison.
16	Full rotating QNM spectrum	Spectral requirements formulated	Evaluate converged axial, polar, and matter branches across spin.
17	Rotating metric and field	BVP and metric asymptotic check formulated	Coupled $h(r)$–$\omega(r)$ exterior solution with regular boundary data.
18	Complete stability	Acceptance conditions formulated	Hyperbolic principal symbol and bounded coupled perturbation evolution.
19	Prospective prediction	Formulated mathematically	Timestamped numerical prediction protocol for a new EHT or gravitational-wave observation.
20	Alternative models	Formulated mathematically	Common comparison with Kerr, Bardeen, Hayward, and Dymnikova metrics.
21	Full Bayesian GW analysis	Likelihood protocol formulated	Strain-data posterior for γ, nuisance marginalization, and evidence relative to GR waveforms specified.
22	Formation by collapse	Formulated mathematically	Constraint-satisfying dynamical-collapse equations and endpoint criterion specified.
23	Quantum corrections	Backreaction problem formulated	Renormalized stress tensor and semiclassical backreaction equation specified.

29 Conclusion

The SSQC model is proposed as a new mathematical and physical approach to regular black holes. It changes the central power structure of the original ansatz while preserving asymptotic Schwarzschild behavior. The resulting metric is locally analytic at the center and is generated by an explicit magnetic NED Lagrangian. These are concrete improvements: the action, effective source, local smoothness, and local geodesic extension are no longer unspecified.

The analysis also identifies how rotation and matter stability must be treated without assuming that a Kerr-like substitution solves the field equations. The tested algebraic Kerr-like metric with the magnetic Kerr–Newman-type field does not satisfy the full Einstein–NED system and is therefore replaced by the coupled field-equation construction. At the representative spin $a_K = 0.20R_s$, the exact difference $\Delta - r^2 A_{\rm sc} = a_K^2 = 4.0 \times 10^{-2} R_s^2$ also demonstrates that the static radial identity cannot validate the rotating ansatz. The coupled system (105)–(109) therefore gives a candidate slow-rotation boundary-value framework for $h(r)$ and $\omega(r)$, supplemented by the derived metric recursions $h_1 = h_2 = 0$. The

electromagnetic recursions still require an explicit constitutive specification. It is not claimed here as a solved rotating spacetime.

The reconstructed one-invariant NED has an unusual strong-field regime: the displayed hyperbolicity combination changes sign inside $r_{\rm hyp} = (6/5)^{1/4}a \simeq 1.0466a$. This behavior does not change the finite density, finite curvature, or analytic SSQC geometry. On the subextremal branch, the sign conditions $\mathcal{L}_{\mathcal{F}} > 0$ and $\mathcal{H} > 0$ hold throughout $r > r_+$, while the core sign failure remains behind the outer horizon. This establishes exterior constitutive-sign admissibility, not full exterior stability or the completion of every exterior observable calculation. A beyond-minimal theory remains a candidate for a globally hyperbolic core or a horizonless branch; no explicit globally admissible member or coupled stable spectrum is constructed in this work. The core-scale evaporation law considered in Gate 23 is a conditional geometric scaling, not a self-consistent semiclassical restoration of hyperbolicity.

For observational comparison, the official calibrated 2017 M87 EHT data release is an existing, public interferometric data product [20]. The visibility equations specify how an SSQC forward model can be compared with those data, including gain, variability, plasma, and calibration parameters. The controlled generated examples test parts of the computational pipeline and are distinct from an analysis of the calibrated observations.

The established result is the exact static SSQC background and its effective magnetic source. Rotation, complete stability, reflectivity, maximal causal structure, observational inference, dynamical formation, and semiclassical backreaction remain open calculations governed by the explicit acceptance criteria collected in the validation framework.

Within the status convention used here, Gates 1, 2, and 4 are completed within their stated static or local scope. Gate 3 is completed only for the local extension through the regular center; its maximal global part remains outside the scope of this article. Gates 5–23 specify equations and acceptance criteria for the remaining rotating, dynamical, observational, and semiclassical analyses; they are not presented as completed calculations.

Data and code availability

The official calibrated 2017 M87 EHT release and the public GW150914 H1/L1 strain data are real, publicly available observational products cited in this article [20, 22]. This manuscript does not report a data-level likelihood or posterior from those products. The numerical values for the static background follow from the displayed dimensionless equations. Slow-rotation endpoint values are included only as an integration diagnostic; solution arrays and convergence data are not provided. The four figure files required to compile the manuscript are included with the source package.

Use of AI tools

The author used AI language tools for drafting and calculation assistance. All results were independently verified by the author.

A Direct component verification of the static solution

For completeness, this appendix gives a compact component-level check of the exact result used in Gates 1 and 2. With

$$A_{\rm sc} = 1 - \frac{2m(r)}{r}, \qquad m(r) = \frac{Mr^3}{(r^4 + a^4)^{3/4}}, \tag{158}$$

the required derivatives are

$$m'(r) = \frac{3Ma^4r^2}{(r^4+a^4)^{7/4}}, \tag{159}$$

$$m''(r) = \frac{3Ma^4r(2a^4-5r^4)}{(r^4+a^4)^{11/4}}. \tag{160}$$

For the static line element in Eq. (2), the independent mixed Einstein-tensor components are

$$G^t{}_t = G^r{}_r = -\frac{2m'}{r^2}, \qquad G^\theta{}_\theta = G^\phi{}_\phi = -\frac{m''}{r}. \tag{161}$$

Substituting Eqs. (159) and (160) yields

$$-\frac{G^t{}_t}{8\pi} = \frac{3Ma^4}{4\pi(r^4+a^4)^{7/4}} = \rho, \tag{162}$$

$$\frac{G^r{}_r}{8\pi} = -\rho = p_r, \tag{163}$$

$$\frac{G^\theta{}_\theta}{8\pi} = \frac{3Ma^4(5r^4-2a^4)}{8\pi(r^4+a^4)^{11/4}} = p_t. \tag{164}$$

Thus all independent Einstein equations are satisfied. On the magnetic background, $\mathcal{F} = q_m^2/(2r^4)$, and direct substitution into Eq. (14) gives

$$\mathcal{L}(\mathcal{F}(r)) = \rho(r), \qquad 2\mathcal{F}\mathcal{L}_\mathcal{F} - \mathcal{L} = p_t. \tag{165}$$

Finally,

$$\sqrt{-g}\,\mathcal{L}_\mathcal{F} F^{\theta\phi} = \frac{q_m \mathcal{L}_\mathcal{F}}{r^2} \tag{166}$$

has no angular dependence, while the radial and temporal components vanish. Therefore $\nabla_\mu(\mathcal{L}_\mathcal{F} F^{\mu\nu}) = 0$, completing the static Einstein–NED verification.

B Constitutive coefficients for slow rotation

For the slow-rotation system in Eqs. (108)–(109), use the parity-even two-invariant law $\mathcal{L} = \mathcal{L}(\mathcal{S}, \mathcal{P}_\mathrm{K}^2)$ and evaluate its derivatives on the magnetic background $\mathcal{S} = q_m^2/r^4$, $\mathcal{P}_\mathrm{K} = 0$. Define

$$\mathcal{L}_\mathcal{S} = \left.\frac{\partial \mathcal{L}}{\partial \mathcal{S}}\right|_{\mathcal{P}_\mathrm{K}=0}, \qquad \mathcal{L}_{\mathcal{P}_\mathrm{K}^2} = \left.\frac{\partial \mathcal{L}}{\partial(\mathcal{P}_\mathrm{K}^2)}\right|_{\mathcal{P}_\mathrm{K}=0}, \tag{167}$$

where a prime below denotes the total derivative with respect to r along this background. In the conventions of Ref. [13], with $f = A_\mathrm{sc}$ and $p = q_m$, the coefficients are

$$\mathfrak{A}(r) = A_\mathrm{sc}\left(r^6\mathcal{L}_\mathcal{S} - 4q_m^2 r^2 \mathcal{L}_{\mathcal{P}_\mathrm{K}^2}\right), \tag{168}$$

$$\mathfrak{B}(r) = rA_\mathrm{sc}\left(r^5\mathcal{L}'_\mathcal{S} - 2r^4\mathcal{L}_\mathcal{S} - 4q_m^2 r\mathcal{L}'_{\mathcal{P}_\mathrm{K}^2} + 24q_m^2\mathcal{L}_{\mathcal{P}_\mathrm{K}^2}\right), \tag{169}$$

$$\mathfrak{C}(r) = A_\mathrm{sc}\left(8q_m^2 r\mathcal{L}'_{\mathcal{P}_\mathrm{K}^2} - 2r^5\mathcal{L}'_\mathcal{S} + 2r^4\mathcal{L}_\mathcal{S} - 40q_m^2\mathcal{L}_{\mathcal{P}_\mathrm{K}^2}\right) - 2r^4\mathcal{L}_\mathcal{S}. \tag{170}$$

The static one-invariant reconstruction fixes $\mathcal{L}_\mathcal{S}$ on this magnetic background, but it does not determine the independent $\mathcal{L}_{\mathcal{P}_\mathrm{K}^2}$ response. Thus these expressions make the radial system explicit without asserting that its matter completion or slow-rotation boundary-value problem has been solved.

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